Spiral Systems Incorporated

# PRODUCT CATALOG

# SPIRAL PIPE AND FITTINGS

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# SINGLE WALL ROUND
Stock Pipe and Fittings

## Spiral Systems Stock Galvanized Gauges

| Diameter | Spiral Pipe Gauge | Fitting Gauge |
|---|---|---|
| 03”-09” | 26 | 22 |
| 10”-18” | 26 | 24 |
| 20” | 24 | 22 |
| 22” | 24 | 22 |
| 24” | 24 | 22 |

- All pipe is stocked in 10’-0” lengths.
- Even and odd sizes available 3” through 9”, even sizes only available 10” through 24”.
- All stock fitting seams are spot welded and sealed with high-pressure duct sealant.
- All stock pipe and fittings are fabricated from G-90 or G-60 galvanized sheet steel meeting ASTM A-653 standards.
- All stock elbows have a centerline radius equal to 1-1/2 times the diameter.
- 3” thru 9” elbows are die-formed elbows, elbows 10” and larger are gored in accordance with SMACNA, 1995.
- Stocking items include: spiral pipe, 45 and 90 degree elbows, couplings, reducers, end caps/plugs, bell mouths, and start collars.
- Stock pipe and fittings are rated for (+) positive 10” W.G. and (-) negative 2” W.G. per SMACNA, 1995.

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# SINGLE WALL ROUND
Positive Pressure Gauges

## POSITIVE PRESSURE
## SPIRAL DUCT & FITTING CONSTRUCTION STANDARDS

- Galvanized
- Paint Grade
- PVC Coated
- Stainless Steel

| | +2" W.G. | | +4" W.G. | | +10" W.G. | |
|---|---|---|---|---|---|---|
| Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" - 6" | 26 | 26 | 26 | 26 | 26 | 26 |
| 7" - 8" | 26 | 26 | 26 | 26 | 26 | 26 |
| 9" - 10" | 26 | 26 | 26 | 26 | 26 | 26 |
| 11" - 12" | 26 | 26 | 26 | 26 | 26 | 24 |
| 13" - 14" | 26 | 26 | 26 | 24 | 26 | 24 |
| 15" - 16" | 26 | 24 | 26 | 24 | 24 | 22 |
| 17" - 18" | 26 | 24 | 24 | 24 | 24 | 22 |
| 19" - 26" | 26 | 24 | 24 | 22 | 24 | 22 |
| 27" - 36" | 24 | 22 | 22 | 20 | 22 | 20 |
| 37" - 50" | 22 | 20 | 20 | 20 | 20 | 20 |
| 51" - 60" | 20 | 18 | 18 | 18 | 18 | 18 |
| 61" - 84" | 18 | 16 | 18 | 16 | 18 | 16 |

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# SINGLE WALL ROUND
## Negative Pressure Gauges

## NEGATIVE PRESSURE
## SPIRAL DUCT & FITTING CONSTRUCTION STANDARDS

- Galvanized
- Paint Grade
- PVC Coated
- Stainless Steel

| | - 2" W.G. | | - 4" W.G. | | - 10" W.G. | |
|---|---|---|---|---|---|---|
| Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" - 8" | 26 | 26 | 26 | 26 | 26 | 26 |
| 9" | 26 | 26 | 26 | 26 | 26 | 24 |
| 10" | 26 | 26 | 26 | 26 | 26 | 22 |
| 11" - 12" | 26 | 26 | 26 | 24 | 24 | 22 |
| 13" | 26 | 26 | 26 | 24 | 24 | 20 |
| 14" | 26 | 24 | 24 | 22 | 24 | 20 |
| 15" | 26 | 24 | 24 | 22 | 22 | 20 |
| 16" | 26 | 24 | 24 | 22 | 22 | 18 |
| 17" | 26 | 24 | 24 | 20 | 22 | 18 |
| 18" - 19" | 24 | 22 | 24 | 20 | 22 | 18 |
| 20" | 24 | 22 | 22 | 20 | 22 | 16 |
| 22" | 24 | 20 | 22 | 18 | 22 | 16 |
| 24" | 22 | 20 | 22 | 18 | 20 | 16 |
| 26" | 22 | 20 | 20 | 18 | 20 | 18R4 |
| 28" - 30" | 22 | 18 | 20 | 16 | 18 | 16R4 |
| 32" | 20 | 18 | 20 | 16 | 18 | 16R4 |
| 34" | 20 | 18 | 20 | 20R6 | 18 | 16R4 |
| 36" | 20 | 16 | 20 | 20R6 | 18 | 16R4 |
| 38" - 42" | 20 | 16 | 18 | 18R6 | 18R12 | |
| 44" - 48" | Consult factory for gauge requirements | | | | | |

R = Reinforcement Ring (Angle Ring) R12=12' centers, R6=6' centers, R4=4' centers.

# SINGLE WALL ROUND SPIRAL DUCT CONSTRUCTION STANDARDS

- **Aluminum 3003-H14**

| | +2" W.G. | | -2" W.G. | |
|---|---|---|---|---|
| **Diameter** | **Spiral Pipe** | **Fittings** | **Spiral Pipe** | **Fittings** |
| 3" - 8" | .025" | .025" | .032" | .032" |
| 9" - 14" | .025" | .025" | .032" | .032" |
| 15" - 26" | .032" | .032" | .040" | .040" |
| 27" - 36" | .040" | .040" | .050" | .050" |
| 37" - 50" | .050" | .050" | .063" | .063" |
| 51" - 60" | .063" | .063" | .071" | .071" |
| 61" - 84" | .063" | .063" | .090" | .090" |

- **Underground PVC Coated Spiral Duct Gauges**

| | +/- 4" W.G. | | Load Specifications | |
|---|---|---|---|---|
| **Diameter** | **Spiral Pipe** | **Fittings** | **Diameter** | **Lbs/ lin. ft.** |
| 3" - 8" | 26 | 24 | 3" - 8" | 400 |
| 9" - 14" | 24 | 22 | 9" - 12" | 600 |
| 15" - 26" | 22 | 20 | 13" - 26" | 1800 |
| 27" - 36" | 20 | 18 | | |
| 37" - 50" | 18 | 16 | | |

## Spiral Pipe and Fittings
## Material & Fabrication Options

| Material Options | Gauges | Fitting Construction |
|---|---|---|
| **Galvanized Steel** G90 – G60, ASTM 527-1 | ▪ Pipe: 26 GA to 16 GA<br>▪ Fittings: 24 GA to 16 GA | ▪ Spot welded and sealed<br>▪ Continuously welded |
| **Paint Grade** | ▪ Pipe: 26 GA to 16 GA<br>▪ Fittings: 24 GA to 16 GA | ▪ Spot welded and sealed<br>▪ Continuously welded |
| **Aluminum** 3003-H14 | ▪ Pipe: .025" to .063"<br>▪ Fittings: .032" to .090" | ▪ Spot welded and sealed<br>▪ Continuously welded |
| **PVC Coated (4 mil x 1 mil)** (Polyvinyl-Chloride-Coated Galvanized Steel) | ▪ Pipe: 26 GA to 16 GA<br>▪ Fittings: 24 GA to 16 GA | ▪ Mechanically Fastened & Sealed |
| **PVC Coated (4 mil x 4 mil)** (Polyvinyl-Chloride-Coated Galvanized Steel) | ▪ Pipe: 26 GA to 16 GA<br>▪ Fittings: 24 GA to 16 GA | ▪ Mechanically Fastened & Sealed |
| **Stainless Steel** (Type 304-2B or 316-2B) | ▪ Pipe: 26 GA to 18 GA<br>▪ Fittings: 24 GA to 16 GA | ▪ Spot welded and sealed<br>▪ Continuously welded |

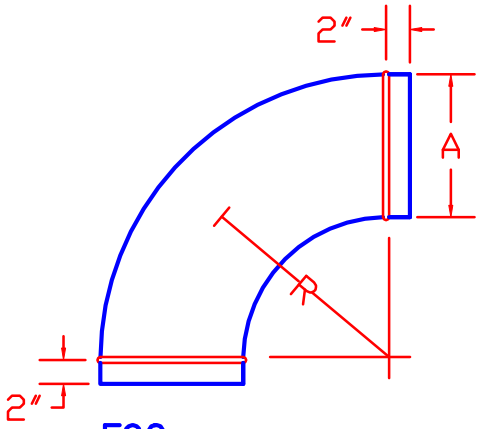
- Standard fittings are spot welded and sealed with high-pressure water based duct sealant.
- Standard single wall pipe and fitting construction is slip joint. All pipe being female and all fittings being male.
- Fully welded fittings are available. (Except PVC coated)
- Pipe and fittings are available from 3" round to 96" round.
- Standard pipe length is 10'-0", other length are available as required.

All fittings and spiral pipe are fabricated per SMACNA HVAC Duct Construction Standards, Second Edition, 1995.

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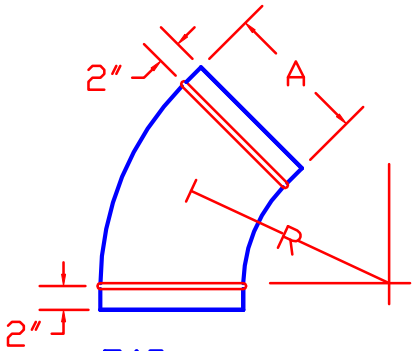
# SINGLE WALL ROUND

## Dieform Elbows




E90
90° DIEFORM ELBOW
STANDARD SIZES: 3" THRU 9"

STANDARD CENTER LINE RADIUS
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A

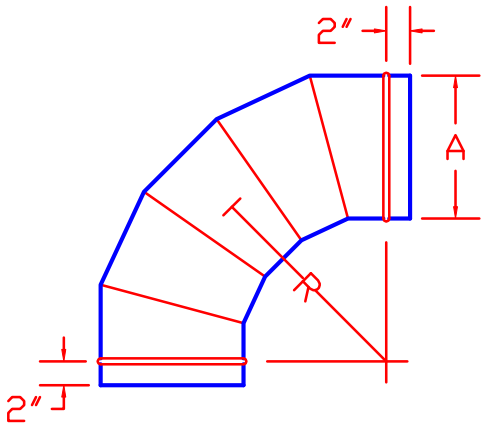



E45
45° DIEFORM ELBOW
STANDARD SIZES: 3" THRU 9"

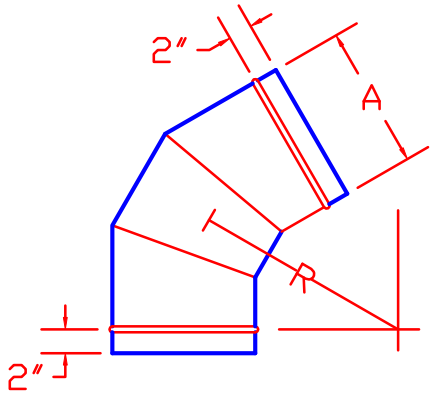
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# SINGLE WALL ROUND

## Gored Elbows

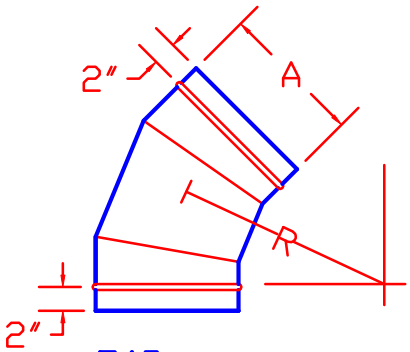



E90
90° 5–PIECE ELBOW

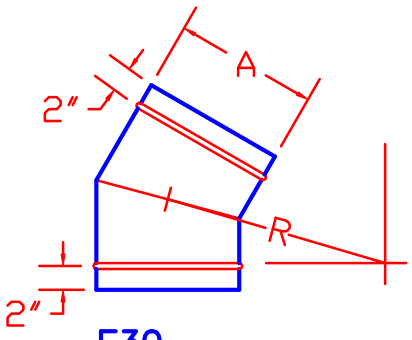



E60
60° 3–PIECE ELBOW

STANDARD CENTER LINE RADIUS
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A




E45
45° 3–PIECE ELBOW

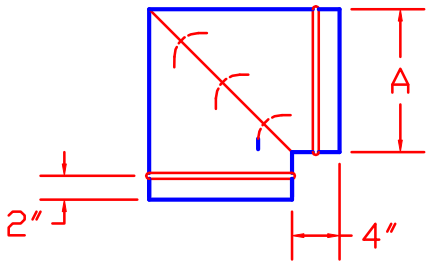



E30
30° 2–PIECE ELBOW

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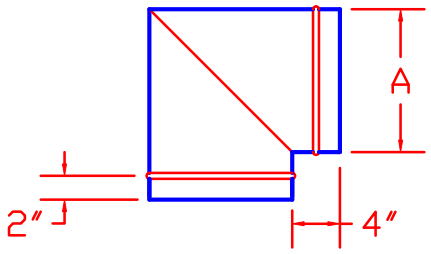
# SINGLE WALL ROUND

## Mitered Elbows

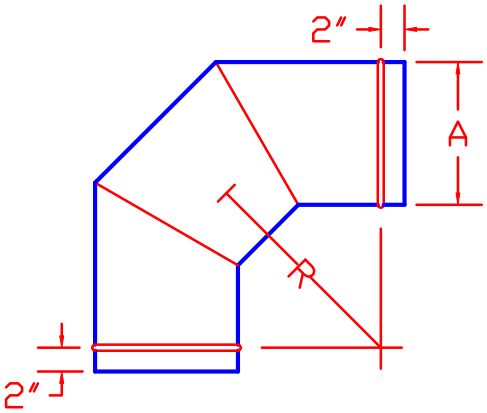



MEV90
90° 2–PIECE MITERED ELBOW W/ VANES

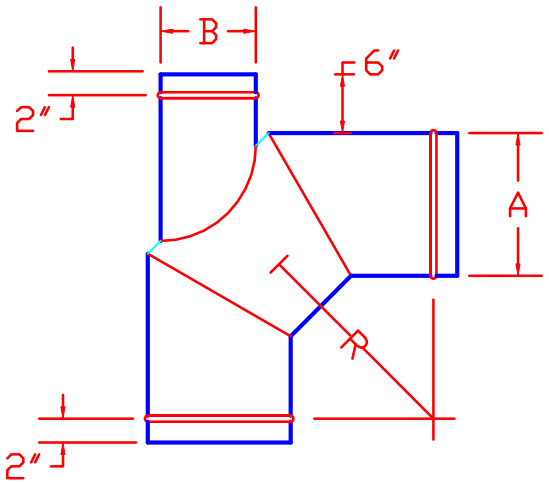
| Dia. (A) | # of Vanes |
|---|---|
| 8" – 9" | 2 |
| 10" – 14" | 3 |
| 15" – 19" | 4 |
| 20" – 60" | 5 |




ME90
90° 2–PIECE MITERED ELBOW




3P90
90° 3–PIECE ELBOW



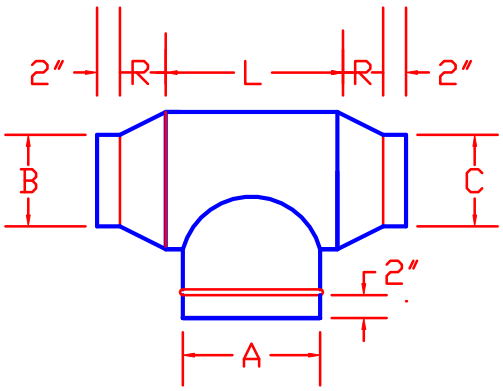

3P90H
90° 3–PIECE ELBOW

STANDARD CENTER LINE
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A

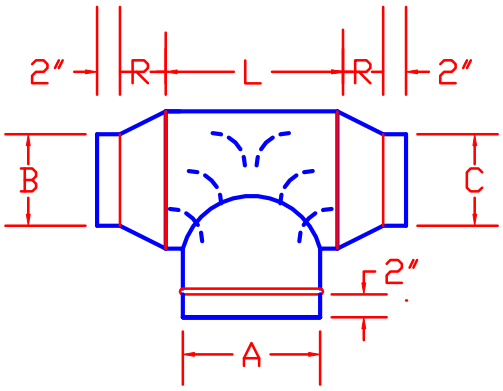
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# SINGLE WALL ROUND

## Bullhead Tees




BT90
BULLHEAD TEE 90°
L= A + 3"
R= Refer to reducer drawing.

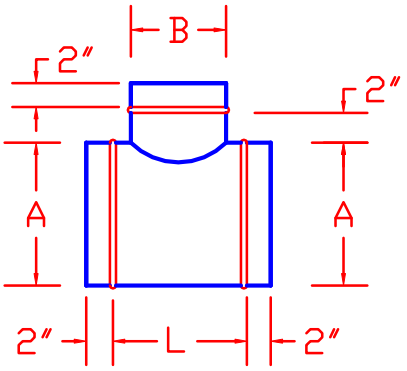



BTV90
BULLHEAD TEE 90° W/ VANES
L= A + 3"
R= Refer to reducer drawing.

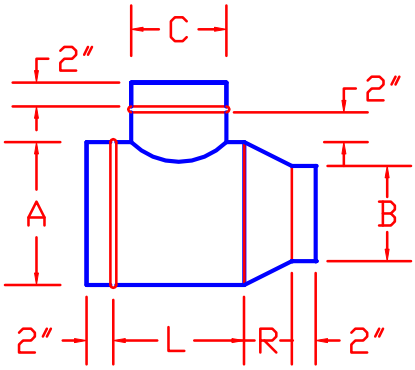
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# SINGLE WALL ROUND

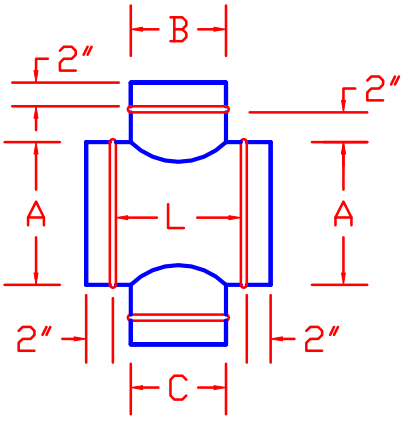
## Tees and Crosses



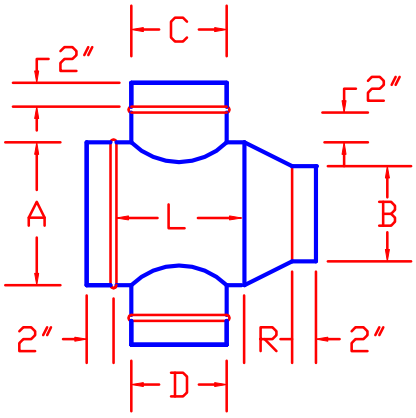

T90
TEE 90°
L= B + 3"




RT90
REDUCING TEE 90°
L= C + 3"
R= Refer to reducer drawing.




C180
CROSS 180°
L= Largest Tap + 3"

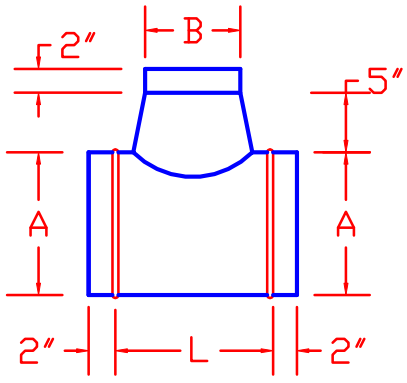



RC180
REDUCING CROSS 180°
L= Largest Tap + 3"
R= Refer to reducer drawing.

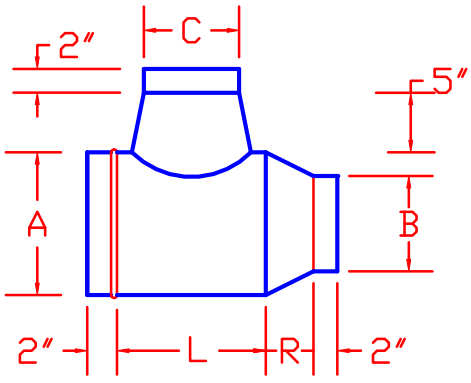
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# SINGLE WALL ROUND

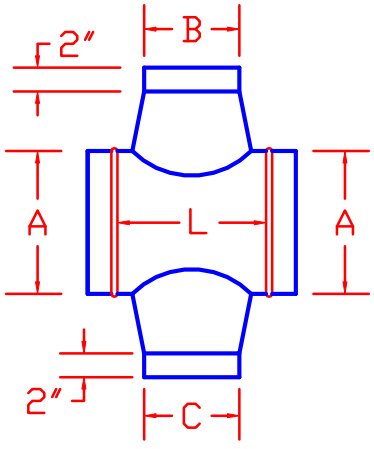
## Conical Tees and Conical Crosses



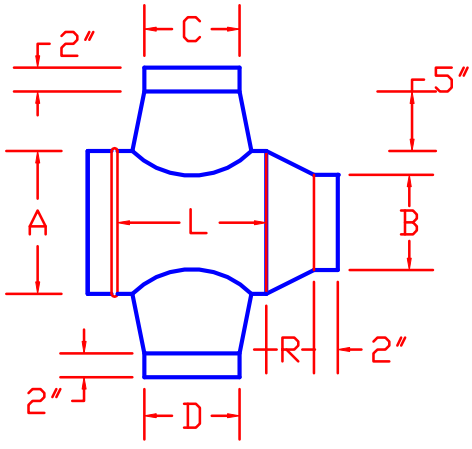

CT90
CONICAL TEE 90°
L= B + 5"




RCT90
REDUCING CONICAL TEE 90°
L= C + 5"
R= Refer to reducer drawing.




CC180
CONICAL CROSS 180°
L= B + 5", B>C

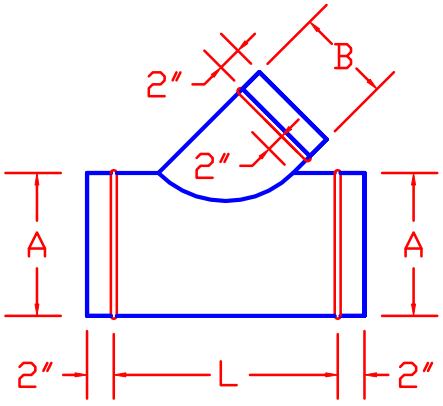



RCC180
REDUCING CONICAL CROSS 180°
L= C + 5",C>D
R= Refer to reducer drawing.

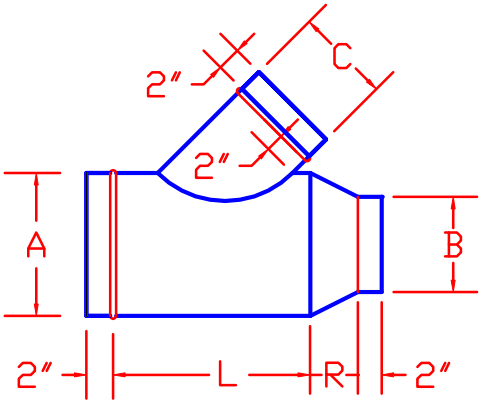
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# SINGLE WALL ROUND

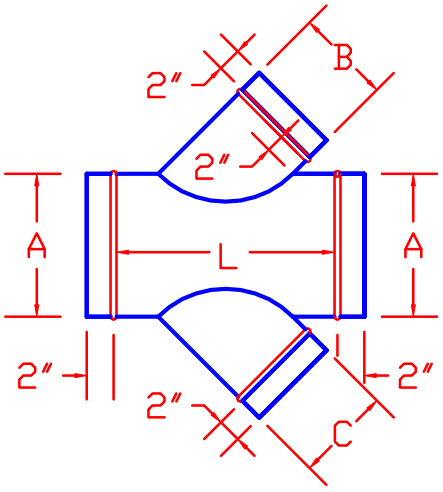
## Laterals



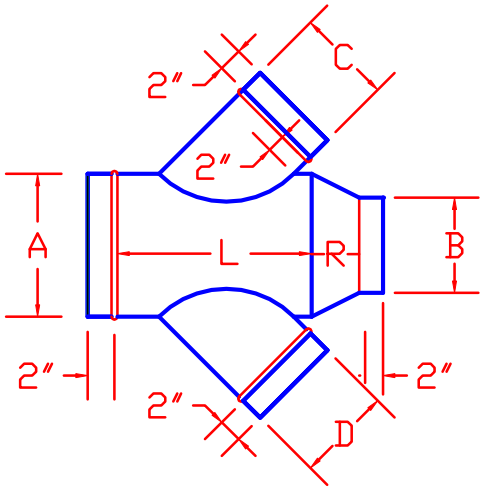

L45
LATERAL 45°
L= B x 1.414 + 3”




RL45
REDUCING LATERAL 45°
L= C x 1.414 + 3”
R= Refer to reducer drawing.




LC45
LATERAL CROSS 45°
L= B x 1.414 + 3”, B>C

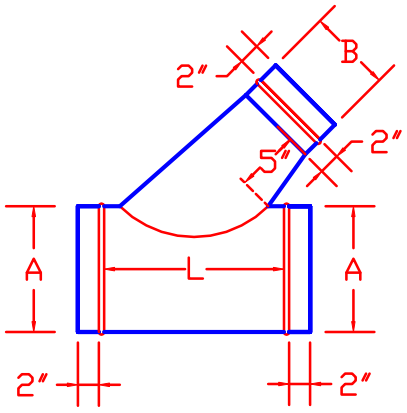



RLC45
REDUCING LATERAL CROSS 45°
L= C x 1.414 + 3”, C>D
R= Refer to reducer drawing.

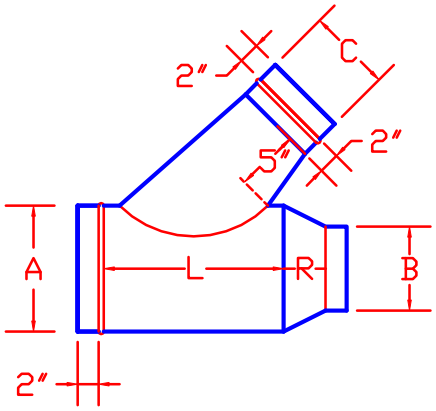
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# SINGLE WALL ROUND

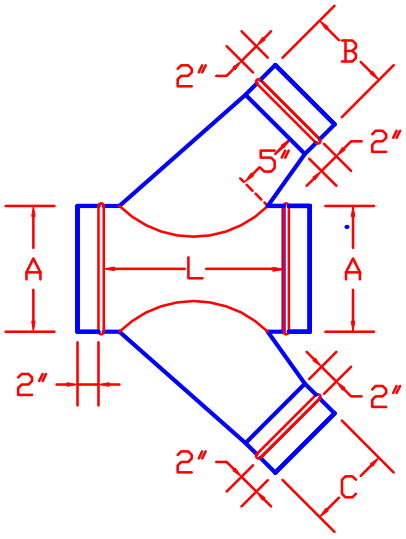
## Conical Laterals



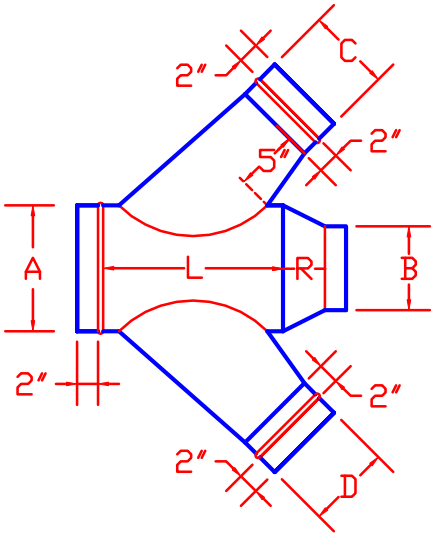

CL45
CONICAL LATERAL 45°
L= B x 1.414 + 5"




CRL45
CONICAL REDUCING LATERAL 45°
L= C x 1.414 + 5"
R= Refer to reducer drawing.




CLC45
CONICAL LATERAL CROSS 45°
L= Largest Tap x 1.414 + 5"

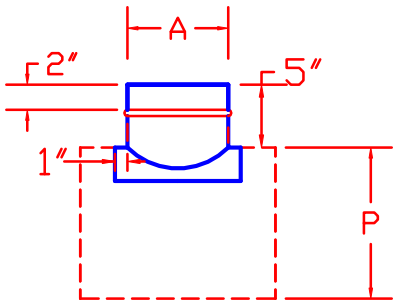



CRLC45
CONICAL REDUCING LATERAL CROSS 45°
L= Largest tap x 1.414 + 5"
R= Refer to reducer drawing.

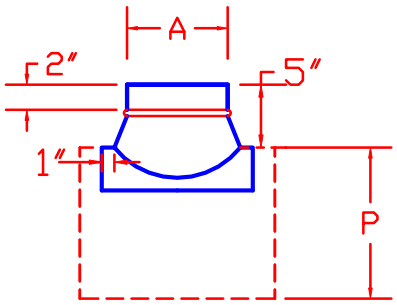
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# SINGLE WALL ROUND

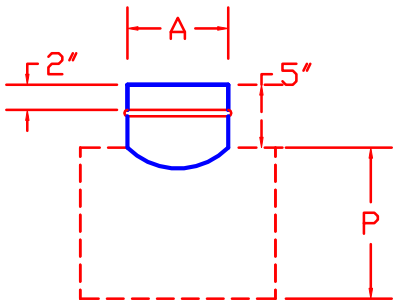
## Straight and Conical Taps



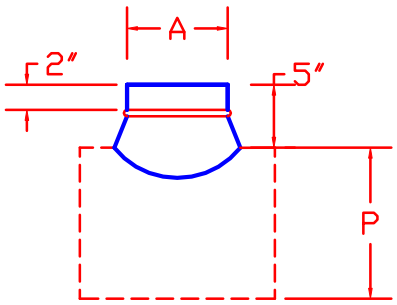

SST90
STRAIGHT SADDLE TAP 90°
TO ORDER LIST A
and P (Pipe Size)
Field Installed




CT90
SADDLE TAP 90°
TO ORDER LIST A
and P (Pipe Size)
Field Installed




ST90
TAP 90°
TO ORDER LIST B
AND P (Pipe Size)
(SADDLE NOT INCLUDED)
Factory or Field Installed

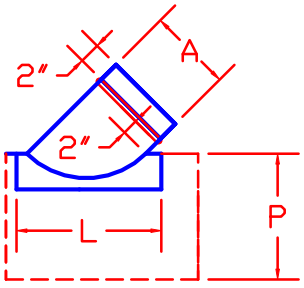



CT90
TAP 90°
TO ORDER LIST A
AND P (Pipe Size)
(SADDLE NOT INCLUDED)
Factory or Field Installed

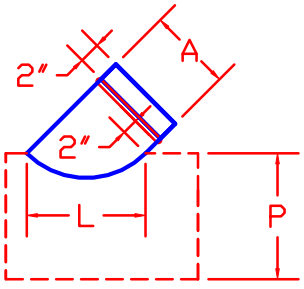
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# SINGLE WALL ROUND

## Lateral Taps




LST45
LATERAL SADDLE TAP 45°
TO ORDER LIST A AND P (Pipe Size)
L= C x 1.414 + 3"

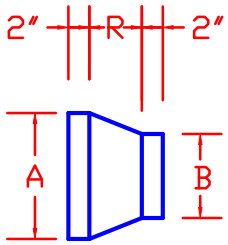



LT45
LATERAL TAP 45°
TO ORDER LIST A AND P (Pipe Size)
L= C x 1.414
(SADDLE NOT INCLUDED)
FACTORY OR FIELD INSTALLED

Spiral Systems Incorporated

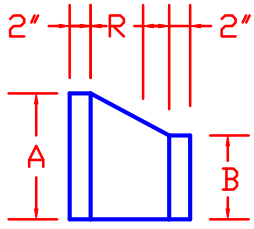
# SINGLE WALL ROUND

## Concentric and Eccentric Reducer




CR
CONCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 4" | 4 |
| 5" THRU 8" | 8 |
| 9" THRU 12" | 12 |
| 13" THRU 16" | 16 |



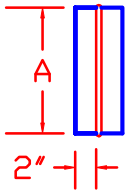

ER
ECCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 3" | 6 |
| (A – B) X 1.5 = | R |

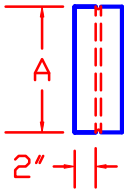
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# SINGLE WALL ROUND

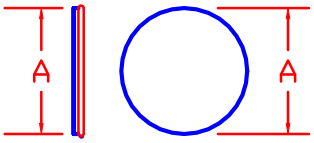
## Pipe Coupling and Fitting Coupling, End Cap, Start Collar, End Cap/Plug Bellmouth



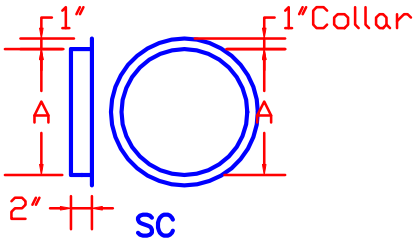

PC
PIPE COUPLING
PIPE TO PIPE CONNECTIONS



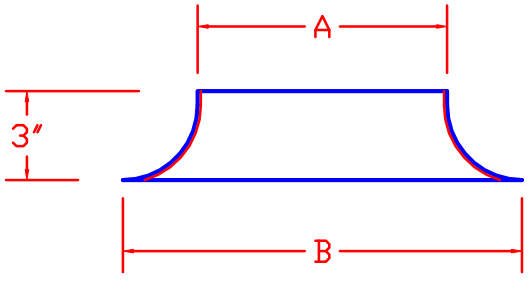

FC
FITTING COUPLING
FITTING TO FITTING CONNECTIONS




EC
END CAP




SC
START COLLAR

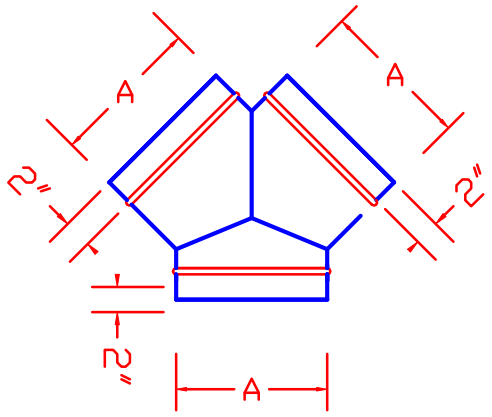



BM
BELL MOUTH
TO ORDER LIST A
B EQUALS A + 4"

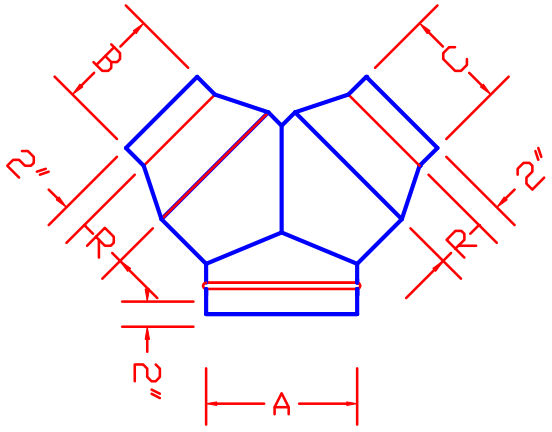
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# SINGLE WALL ROUND

## Wye and Reducing Wye




**Y45**
**WYE 45°**

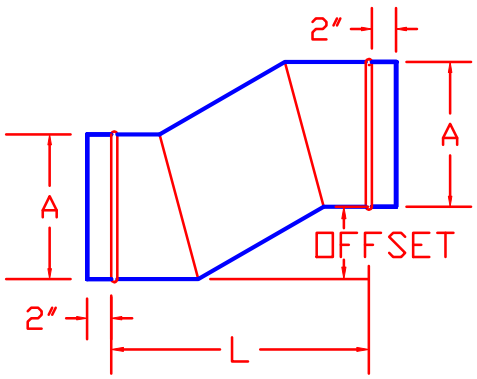



**RY45**
**REDUCING WYE 45°**
**TO ORDER LIST A, B, & C.**
**R= Refer to reducer drawing.**

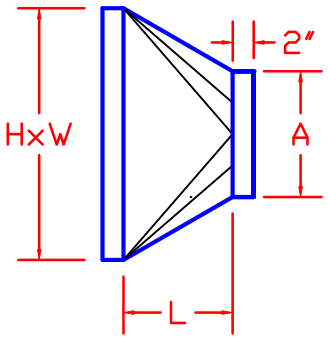
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# SINGLE WALL ROUND

## Offset and Square-to-Round Transition




OS
OFFSET
L= A x 2



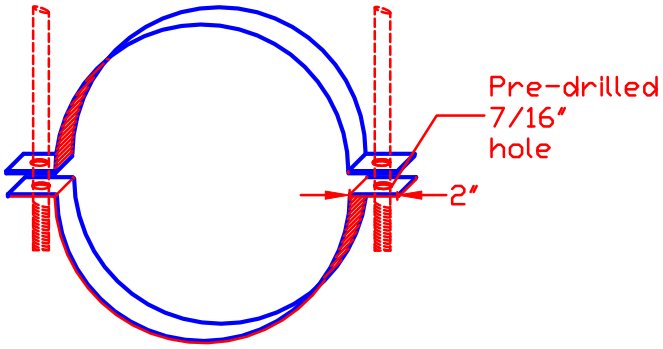

SQ2RD
SQUARE TO ROUND TRANSITION

| A – (LARGEST LENGTH H OR W MINUS ROUND DIAMETER) | L (Inches) |
|---|---|
| 3–8" | 12 |
| 9–14" | 18 |
| 15–26" | 24 |
| 27" and Larger | 36 |

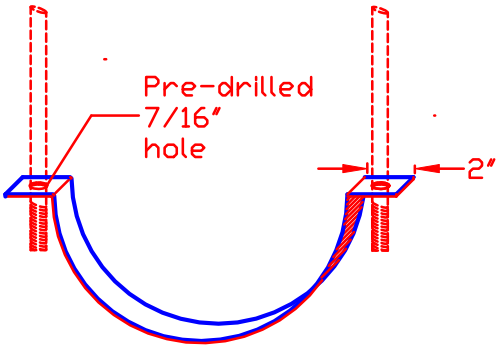
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# SINGLE WALL ROUND

## Clam Shell Hangers




WHOLE CLAM
SHELL HANGER
TO ORDER LIST PIPE SIZE
THREADED ROD NOT INCLUDED.




HALF CLAM
SHELL HANGER
TO ORDER LIST PIPE SIZE
THREADED ROD NOT INCLUDED.

| Pipe Diameter | Width & Gauge |
|---|---|
| 3" THRU 15" | 1" Wide x 20 Ga. |
| 16" THRU 28" | 1" Wide x 18 Ga. |
| 30" THRU 38" | 1" Wide x 16 Ga. |
| 40" THRU 64" | 1-1/2" Wide x 16 Ga. |

Spiral Systems Incorporated

# SINGLE WALL ROUND

## Access Doors / Duct Connectors

## ACCESS DOOR



**Description:**
Solid, leak proof, insulated, access door for round ducts.

Pressure rating: +20" W.G. to –10" W.G.

Service temperature range: -20°F to 200°F.

Options: Non-Insulated, Observation, and High Temperature.

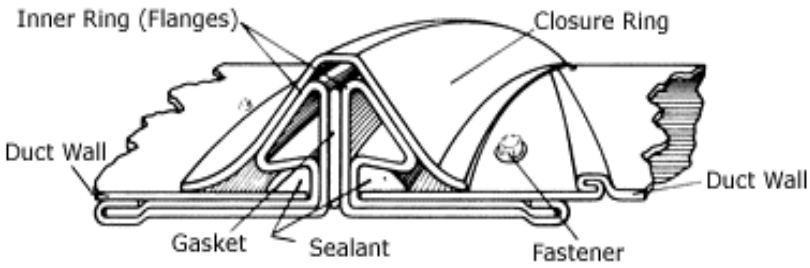
| Door Size | Duct Size |
|---|---|
| 8" x 4" | 5" - 9" |
| 12" x 8" | 10" - 20" |
| 16" x 12" | 21" - 36" |
| 24" x 18" | 37" - 64" |

## DUCT CONNECTION FLANGES

**Spiralmate™** by
Ductmate Industries, Inc.

Available Sizes: 10" thru 72"

Factory Installed or Loose for Field Installation



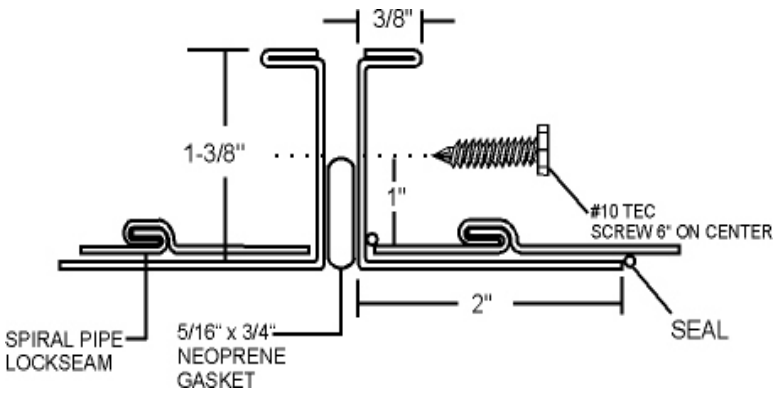

Material Options: Galvanized, Stainless Steel , PVC Coated, Aluminized.

**Accuflange™** by
Accuduct Manufacturing, Inc.

Available Sizes: 10" thru 96"

Factory Installed or Loose for Field Installation.

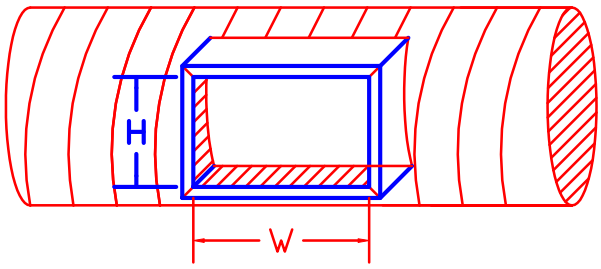
Material Options: Galvanized, Paint Grade, Stainless Steel , PVC Coated, Aluminum.



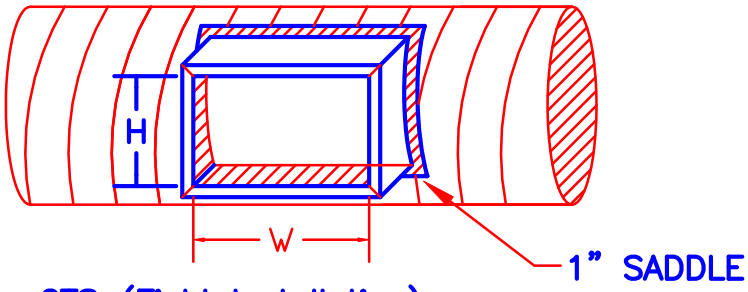
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# SINGLE WALL ROUND

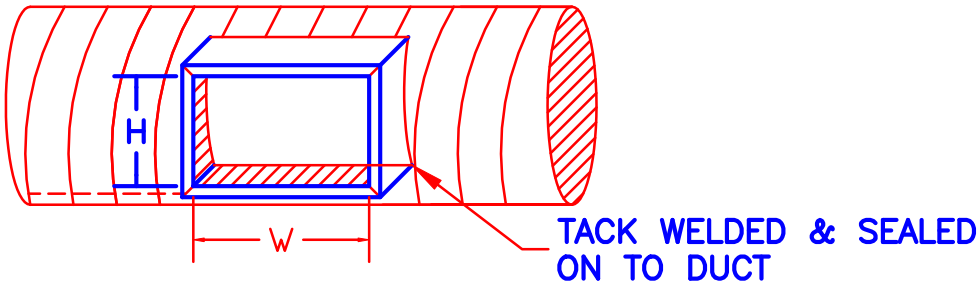
## Grille Taps




GT (Field Installation)
GRILLE TAP (No Saddle)
TO ORDER LIST W, H, & PIPE SIZE.




GTS (Field Installation)
GRILLE TAP with 1" SADDLE INCLUDED
TO ORDER LIST W, H, & PIPE SIZE.

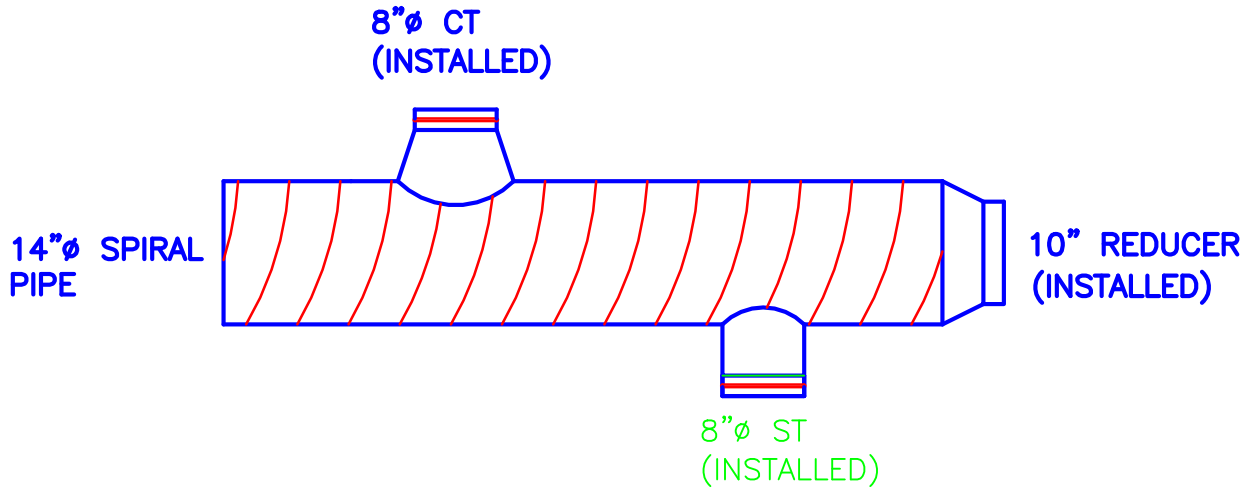


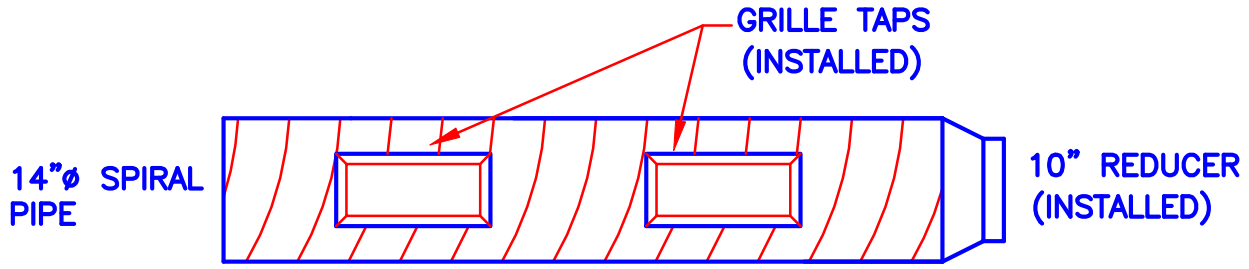

GTI (Factory Installed on Pipe)
GRILLE TAP INSTALLED
TO ORDER LIST W, H, & PIPE SIZE.

Spiral Systems Incorporated

# SINGLE WALL ROUND

## Manifolding




## MANIFOLD SPIRAL PIPE

- REDUCES INSTALLATION COST
  - ELIMINATES A MAJORITY OF THE PIPE / FITTING CONNECTIONS.
  - ROUND SECTIONS AVALIBLE IN LENGTHS UP TO 20'-0".
  - ALL MANIFOLDED SECTIONS ARE NUMBERED AND COLOR-CODED BY FLOOR OR AHU.
  - FACTORY INSTALLED FLANGES ARE AVALIBLE.
- TAPS ARE INSTALLED ON THE PIPE AS SPECIFIED BY THE DRAWING.
- MANIFOLDING REDCES DUCT LEAKAGE BY ELIMINATING THE NUMBER OF JOINTS.
- STANDARD TAPS ARE TACK WELDED AND SEALED.
  (CONTINOUSLY WELDED TAPS ARE AVALIBLE)

Spiral Systems, Inc.

P.O. Box 513, Nesbit, MS  38651

• Phone:  (877) 280-2663 • Fax: (888) 429-3373 •

Spiral Systems Incorporated

# PRODUCT CATALOG

# DOUBLE WALL ROUND SPIRAL PIPE AND FITTINGS

Spiral Systems, Inc.

P.O. Box 513, Nesbit, MS  38651

• Phone:  (877) 280-2663 • Fax: (888) 429-3373 •

Spiral Systems Incorporated

# DOUBLE WALL ROUND
Stock Pipe and Fittings

## Double Wall Round Spiral Pipe and Fittings Construction

**Double Wall Round – Spiral Pipe – Solid / Solid**
Solid / Solid means solid materials are used on both the outer and inner shells.

**Double Wall Round – Spiral Pipe – Solid / Perforated**
Solid / Perforated means solid materials are used on the outer shell and perforated materials on the inner shells.

- Standard double wall pipe and fittings are supplied with 1" thick, 1-1/2# density fiberglass insulation.
- Double wall spiral pipe is available in lengths from 1 to 20 feet. (Specify when ordering).
- Standard double wall fittings are provided with a solid inner liner. Fittings are available with perforated liner when specified.
- Insulation is available in 1", 2", and 3" thickness with 1-1/2# density.
- Standard double wall pipe and fittings are slip joint construction; other connection options are available upon request.
- Pipe and fittings are available in all SMACNA pressure classifications.
- Pipe and fittings can be fabricated for either positive or negative systems.

Spiral Systems Incorporated

# DOUBLE WALL ROUND
## Spiral Pipe and Fittings

## Positive Pressure
## Spiral Duct & Fittings Construction Standards

- Galvanized
- Paint Grade
- Stainless Steel

| | +2" W.G. I.D. | | +2" W.G. O.D. | |
|---|---|---|---|---|
| Inner Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" – 14" | 26 | 26 | 26 | 26 |
| 15" – 26" | 26* | 24 | 26 | 24 |
| 27" – 36" | 24* | 22 | 24 | 22 |
| 37" – 50" | 22* | 20 | 22 | 20 |
| 51" – 60" | 20* | 18 | 20 | 18 |
| 61" – 86" | 18* | 18 | 18 | 16 |

| | +4" W.G. I.D. | | +4" W.G. O.D. | |
|---|---|---|---|---|
| Inner Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" – 13" | 26 | 26 | 26 | 26 |
| 14" – 18" | 26 | 24 | 26 | 24 |
| 19" – 26" | 26* | 24 | 24 | 22 |
| 27" – 36" | 24* | 22 | 22 | 20 |
| 37" – 50" | 22* | 20 | 20 | 20 |
| 51" – 60" | 20* | 18 | 18 | 18 |
| 61" – 86" | 18* | 18 | 18 | 16 |

| | +10" W.G. I.D. | | +10" W.G. O.D. | |
|---|---|---|---|---|
| Inner Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" – 11" | 26 | 26 | 26 | 26 |
| 12" – 15" | 26 | 24 | 26 | 24 |
| 16" – 26" | 26* | 24 | 24 | 22 |
| 27" – 36" | 24* | 22 | 22 | 20 |
| 37" – 50" | 22* | 20 | 20 | 20 |
| 51" – 60" | 20* | 18 | 18 | 18 |
| 61" – 86" | 18* | 18 | 18 | 16 |

* Beaded spiral pipe and fittings.

Notes: a) If longitudinal seam duct is required, the fitting gauge standards will apply.
b) Standard inner liner for all duct is solid galvanized steel, regardless of outer material.
c) Standard inner liner for all fittings is solid galvanized steel, regardless of outer material.

## DOUBLE WALL ROUND SPIRAL DUCT CONSTRUCTION STANDARDS

- Aluminum 3003-H14

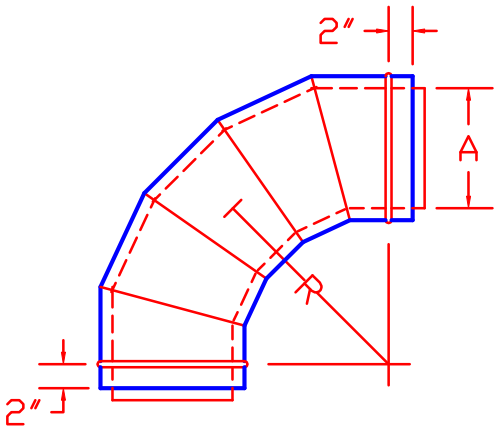
| | I.D. GAUGES | | O.D. GAUGES | |
|---|---|---|---|---|
| | +2" W.G. | | +2" W.G. | |
| Inner Diameter | Spiral Pipe | Fittings | Spiral Pipe | Fittings |
| 3" - 8" | .025" | .032" | .025" | .032" |
| 9" - 14" | .025" | .032" | .025" | .032" |
| 15" - 26" | .032" | .040" | .032" | .040" |
| 27" - 36" | .040" | .050" | .040" | .050" |
| 37" - 50" | .050" | .063" | .050" | .063" |
| 51" - 60" | .063" | .071" | .063" | .071" |
| 61" - 84" | NA | .090" | NA | .090" |

NA – Not avalible or not assigned.

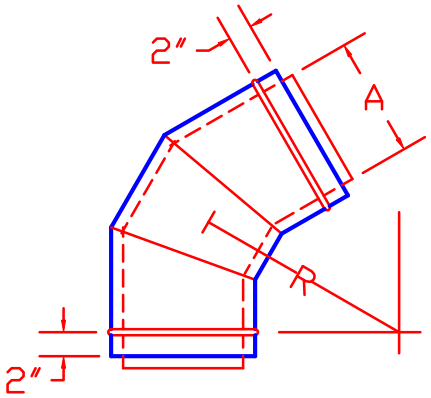
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# DOUBLE WALL ROUND

## Gored Elbows

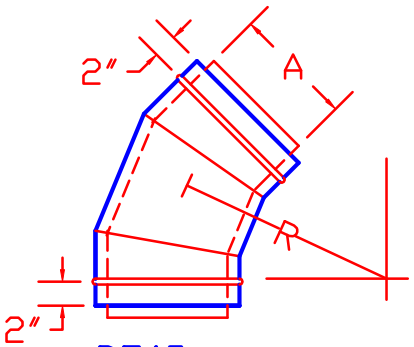



DE90
90° 5–PIECE ELBOW

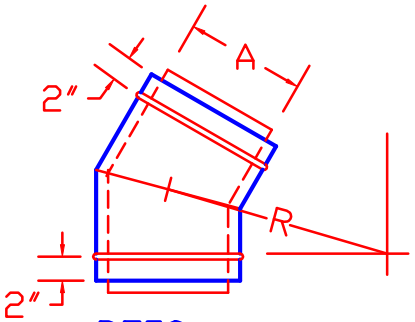



DE60
60° 3–PIECE ELBOW

STANDARD CENTER LINE RADIUS
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A




DE45
45° 3–PIECE ELBOW

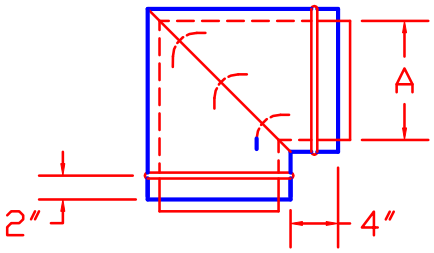



DE30
30° 2–PIECE ELBOW

Spiral Systems Incorporated

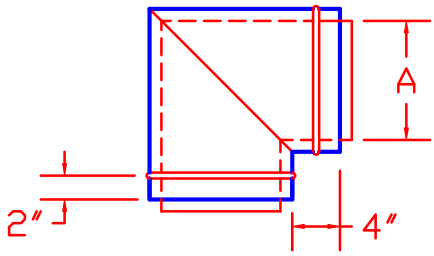
# DOUBLE WALL ROUND

## Mitered Elbows

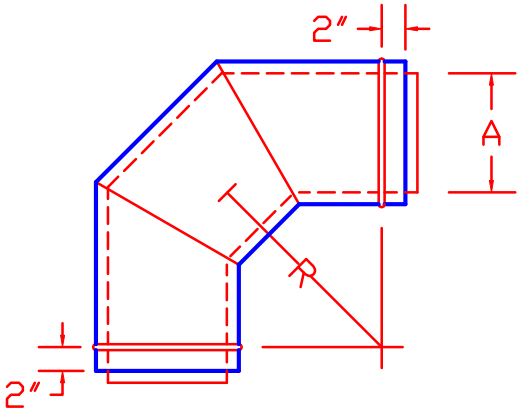



DMEV90
90° 2–PIECE MITERED ELBOW W/ VANES

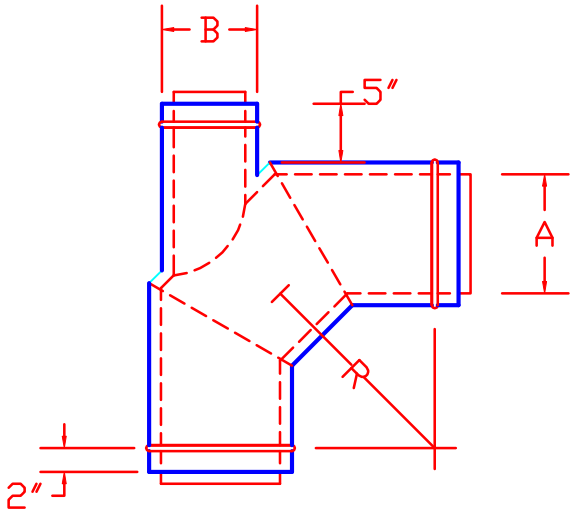
| Dia. (A) | # of Vanes |
|---|---|
| 8" – 9" | 2 |
| 10" – 14" | 3 |
| 15" – 19" | 4 |
| 20" – 60" | 5 |




DME90
90° 2–PIECE MITERED ELBOW




D3P90
90° 3–PIECE ELBOW



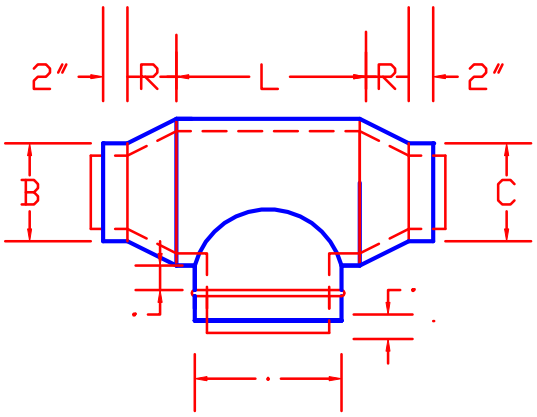

D3P90H
90° 3–PIECE ELBOW

STANDARD CENTER LINE
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A

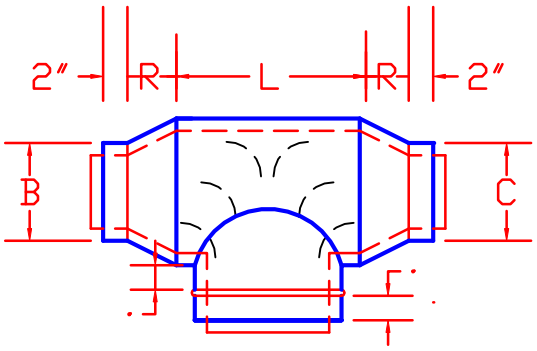
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# DOUBLE WALL ROUND

## Bullhead Tees




DBT90
BULLHEAD TEE 90°
L= A + 3"
R= Refer to reducer drawing.

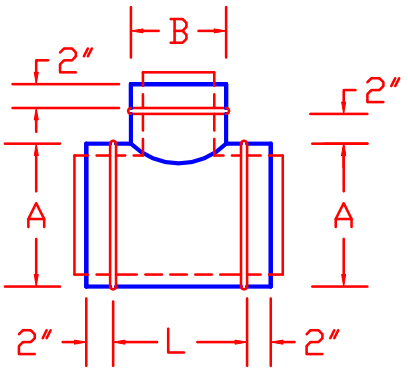



DBTV90
BULLHEAD TEE 90° W/ VANES
L= A + 3"
R= Refer to reducer drawing.

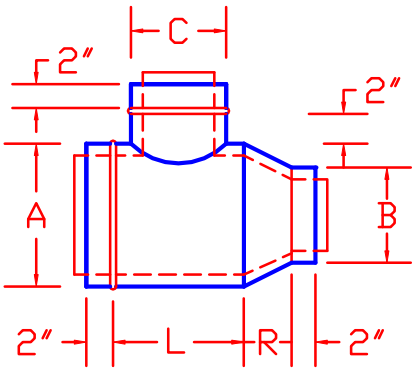
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Systems
Incorporated

# DOUBLE WALL ROUND

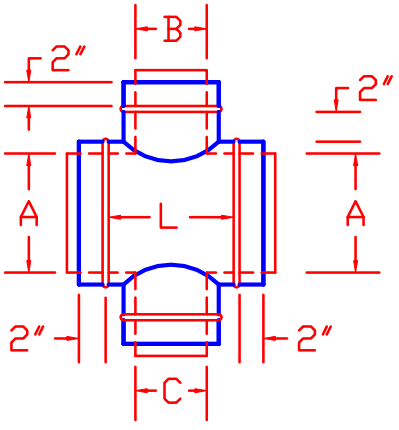
## Tees and Crosses



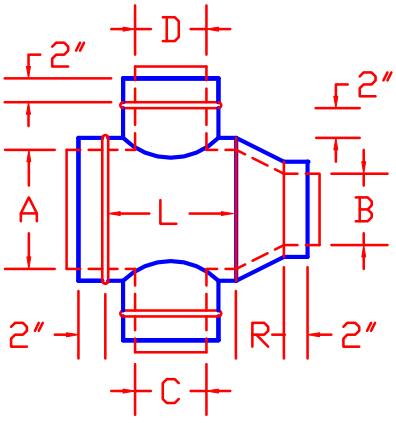

DT90
TEE 90°
L= B+3"




DRT90
REDUCING TEE 90°
L= B+3"
R= Refer to reducer drawing.




DC180
CROSS 180°
L= Largest Tap+3"

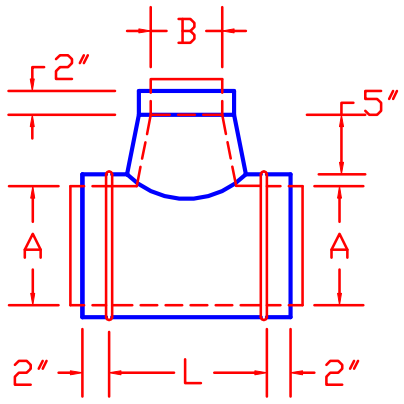



DRC180
REDUCING CROSS 180°
L= Largest Tap+3"
R= Refer to reducer drawing.

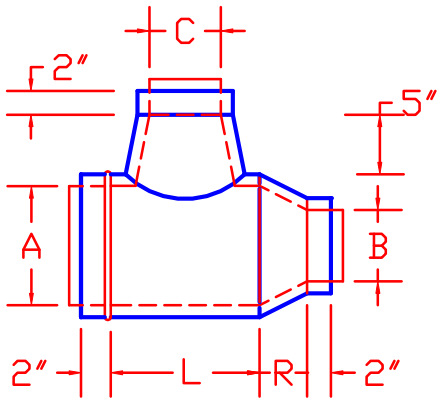
Spiral Systems Incorporated

# DOUBLE WALL ROUND

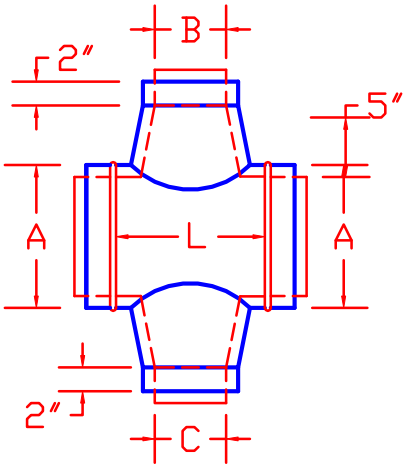
## Conical Tees and Conical Crosses



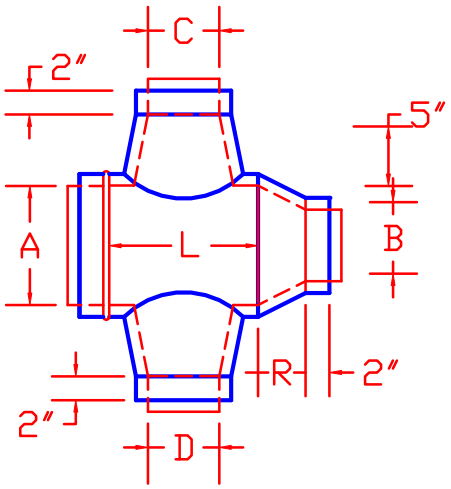

DCT90
CONICAL TEE 90°
L= B +5"




DRCT90
REDUCING CONICAL TEE 90°
L= C+5"
R= Refer to reducer drawing.




DCC180
CONICAL CROSS 180°
L= Largest Tap +5"

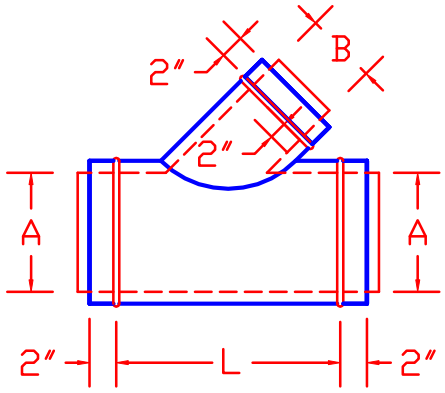



DRCC180
REDUCING CONICAL CROSS 180°
L= Largest Tap+5"
R= Refer to reducer drawing.

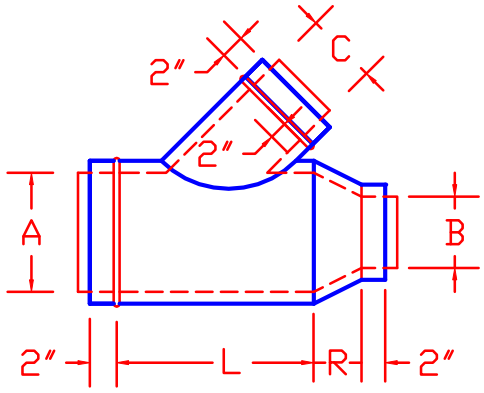
Spiral Systems Incorporated

# DOUBLE WALL ROUND

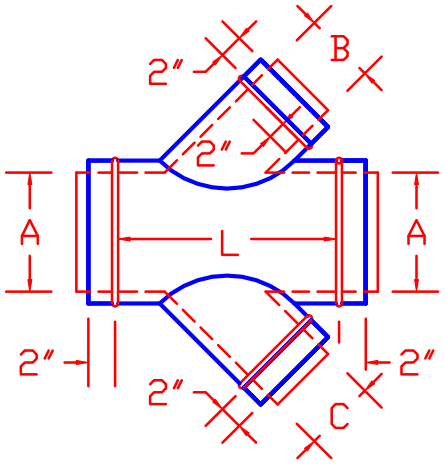
## Laterals



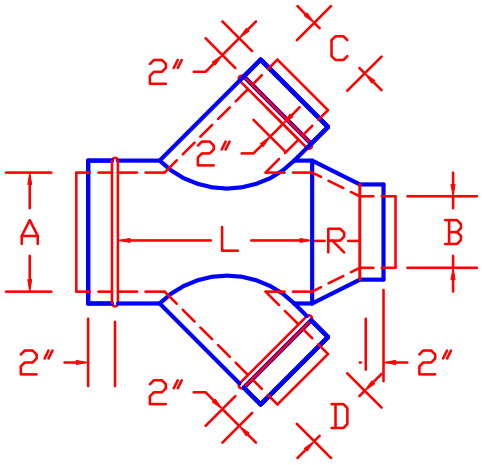

DL45
LATERAL 45°
L= C x 1.414 + 3"




DRL45
REDUCING LATERAL 45°
L= C x 1.414 + 3"
R= Refer to reducer drawing.




DLC45
LATERAL CROSS 45°
L= B x 1.414 + 3", B>C

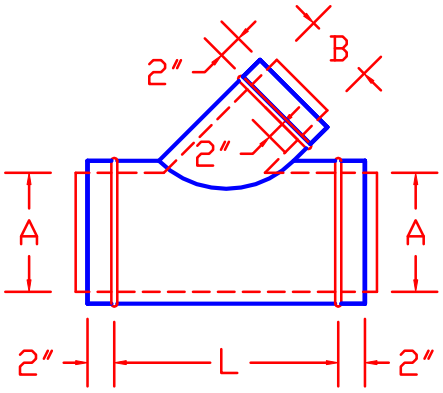



DRLC45
REDUCING LATERAL CROSS 45°
L= C x 1.414 + 3", C>D
R= Refer to reducer drawing.

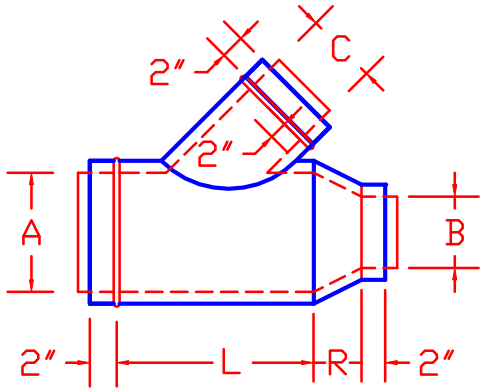
Spiral Systems Incorporated

# DOUBLE WALL ROUND

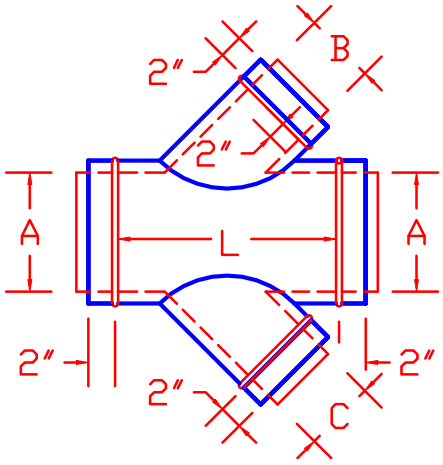
## Conical Laterals



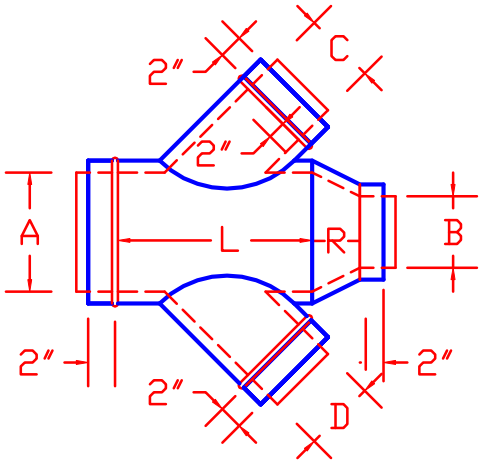

DL45
LATERAL 45°
L= C x 1.414 + 3"




DRL45
REDUCING LATERAL 45°
L= C x 1.414 + 3"
R= Refer to reducer drawing.




DLC45
LATERAL CROSS 45°
L= B x 1.414 + 3", B>C

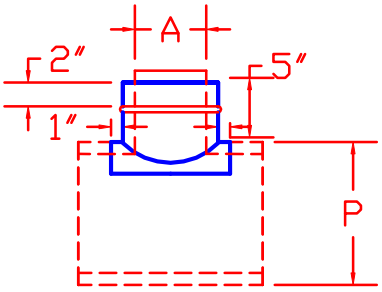



DRLC45
REDUCING LATERAL CROSS 45°
L= C x 1.414 + 3", C>D
R= Refer to reducer drawing.

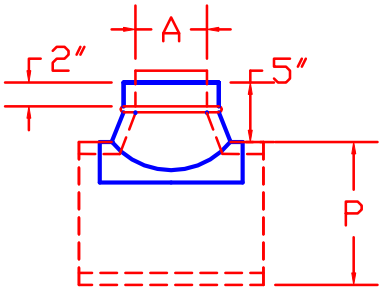
Spiral Systems Incorporated

# DOUBLE WALL ROUND

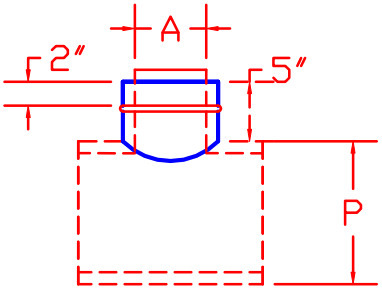
## Straight Taps and Conical Taps



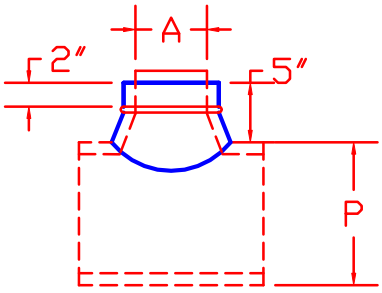

DSST90
STRAIGHT SADDLE TAP 90°
TO ORDER LIST B
and P (Pipe Size)




DCT90
SADDLE TAP 90°
TO ORDER LIST B
and P (Pipe Size)




DST90
TAP 90°
TO ORDER LIST B
AND P (Pipe Size)
(SADDLE NOT INCLUDED)

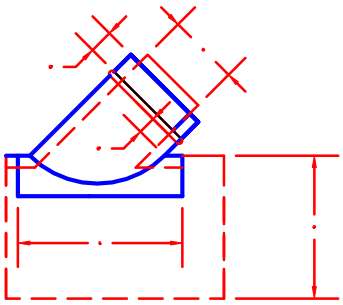



DCT90
TAP 90°
TO ORDER LIST B
AND P (Pipe Size)
(SADDLE NOT INCLUDED)

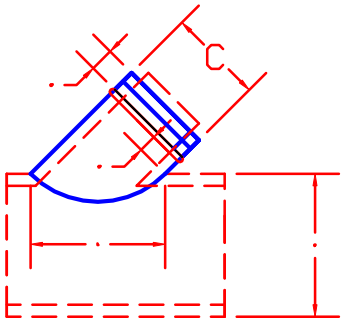
Spiral Systems Incorporated

# DOUBLE WALL ROUND

## Laterals Taps



DLST45
LATERAL SADDLE TAP 45°
TO ORDER LIST A AND P (Pipe Size)
L= C x 1.414 + 3"

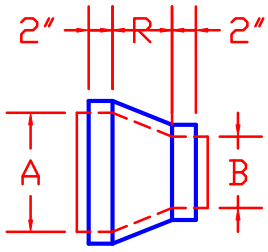



LT45
LATERAL TAP 45°
TO ORDER LIST A AND P (Pipe Size)
L= C x 1.414
(SADDLE NOT INCLUDED)

Spiral Systems Incorporated

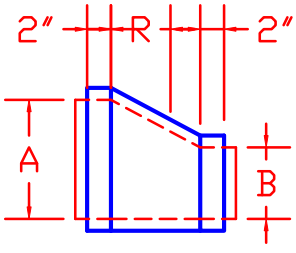
# DOUBLE WALL ROUND

## Concentric and Eccentric Reducer




DCR
DW CONCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 4" | 4 |
| 5" THRU 8" | 8 |
| 9" THRU 12" | 12 |
| 13" THRU 16" | 16 |



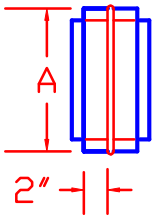

DER
DW ECCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 3" | 6 |
| (A – B) X 1.5 = | R |

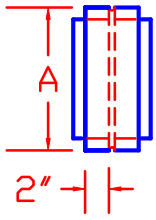
Spiral Systems Incorporated

# DOUBLE WALL ROUND

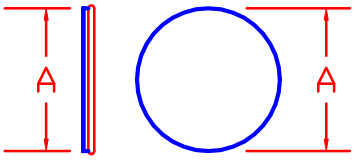
## Pipe and Fitting Coupling, End Cap, and Start Collar




DPC
PIPE COUPLING
PIPE TO PIPE CONNECTIONS




DFC
FITTING COUPLING
FITTING TO FITTING CONNECTIONS

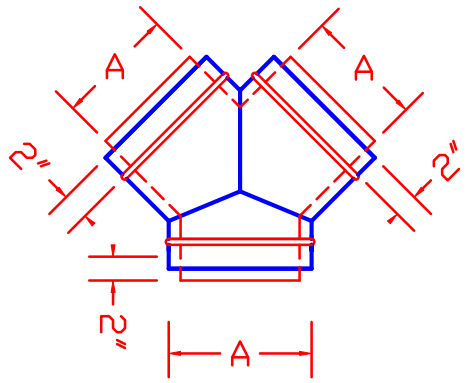



DEC
END CAP

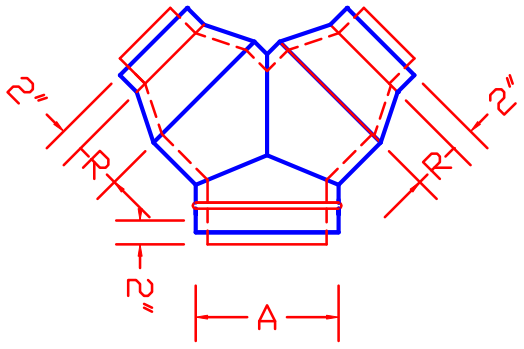
Spiral Systems Incorporated

# DOUBLE WALL ROUND

## Wye and Reducing Wye



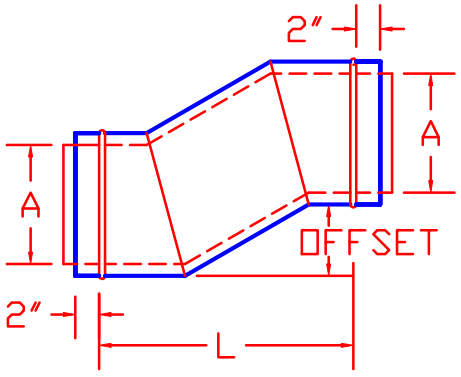

DY45
WYE 45°



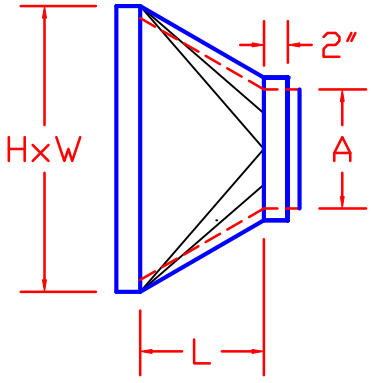

DRY45
REDUCING WYE 45°
R= Refer to reducer drawing.

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# DOUBLE WALL ROUND




DOS
OFFSET
L= A x 2




DSQ2RD
SQUARE TO ROUND TRANSITION

| A – (LARGEST LENGTH H OR W MINUS ROUND DIAMETER) | L (Inches) |
|---|---|

Spiral Systems Incorporated

# DOUBLE WALL ROUND

## Access Doors / Duct Connectors

## ACCESS DOOR



**Description:**
Solid, leak proof, insulated, access door for round ducts.

Pressure rating: +20” W.G. to –10” W.G.

Service temperature range: -20°F to 200°F.

Options: Non-Insulated, Observation, and High Temperature.

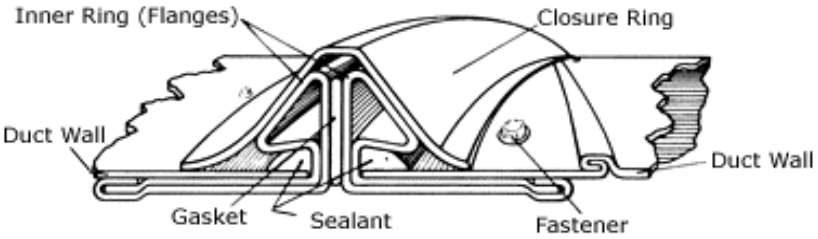
| Door Size | Duct Size |
|---|---|
| 8” x 4” | 5” - 9” |
| 12” x 8” | 10” - 20” |
| 16” x 12” | 21” - 36” |
| 24” x 18” | 37” - 64” |

## DUCT CONNECTION FLANGES

**Spiralmate™** by
Ductmate Industries, Inc.

Available Sizes: 10” thru 72”

Factory Installed or Loose for Field Installation



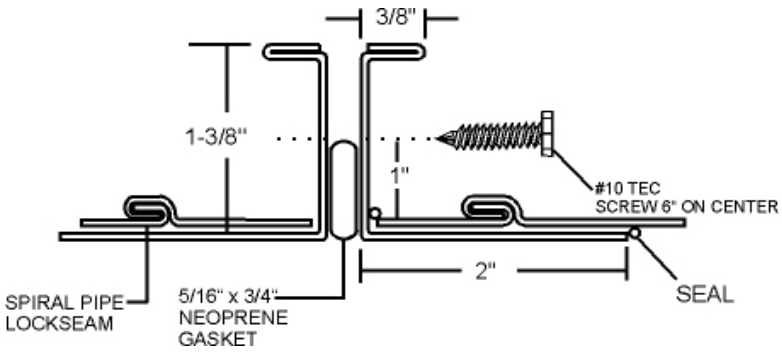

Material Options: Galvanized, Stainless Steel , PVC Coated, Aluminized.

**Accuflange™** by
Accuduct Manufacturing, Inc.

Available Sizes: 10” thru 96”

Factory Installed or Loose for Field Installation.

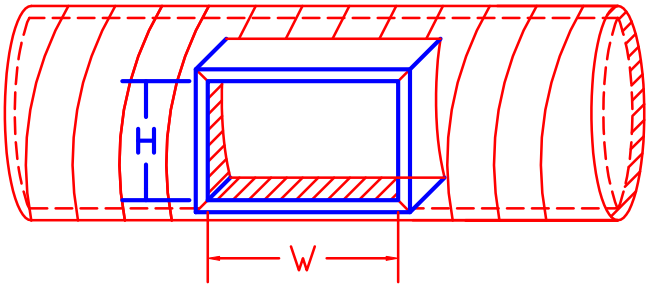
Material Options: Galvanized, Paint Grade, Stainless Steel , PVC Coated, Aluminum.



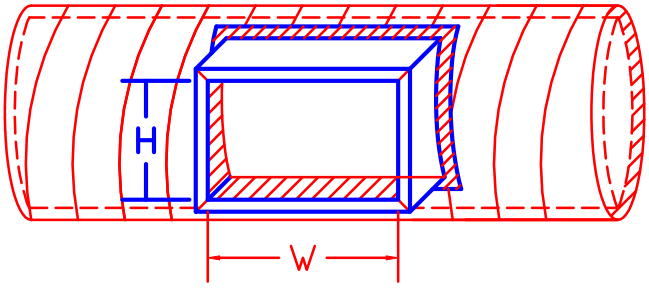
Spiral Systems Incorporated

# DOUBLE WALL ROUND

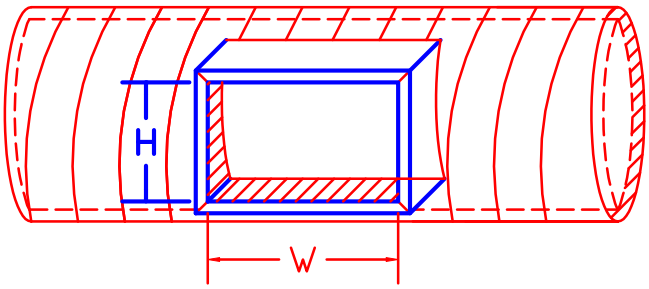
## Grille Taps




GTD (Field Installation)
GRILLE TAP (No Saddle)
TO ORDER LIST W, H, & PIPE SIZE.
R= Refer to reducer drawing.




GTSD (Field Installation)
GRILLE TAP with 1" SADDLE INCLUDED
TO ORDER LIST W, H, & PIPE SIZE.
R= Refer to reducer drawing.

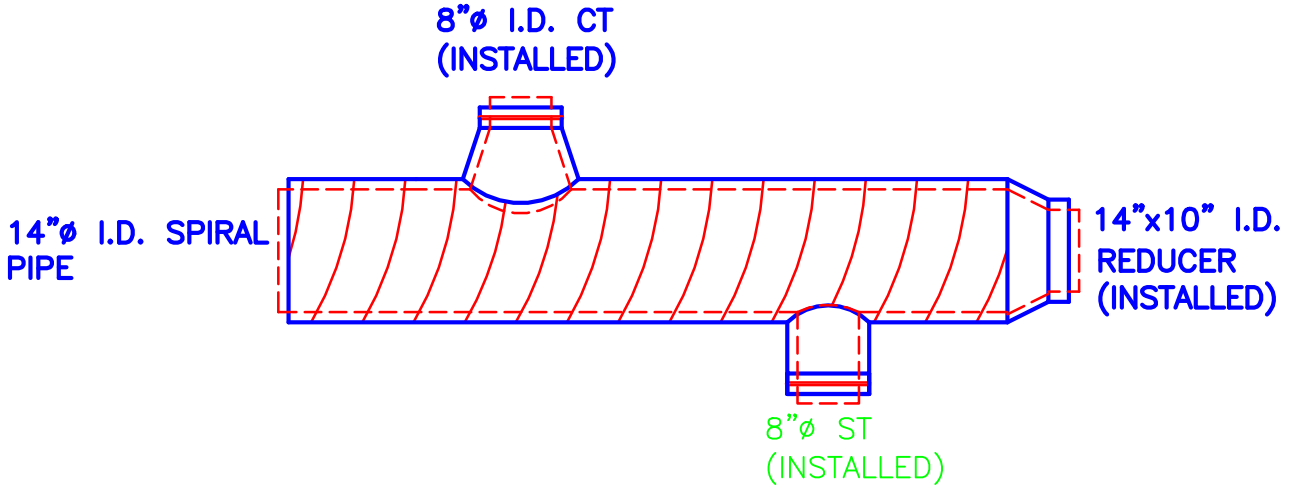


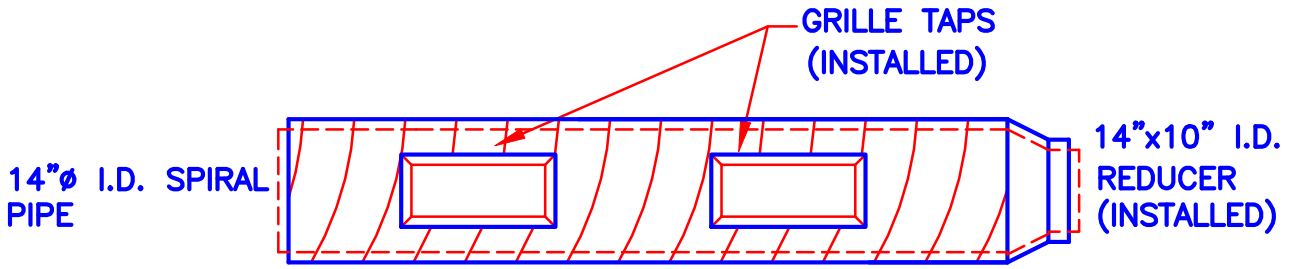

GTID (Factory Installed on Pipe)
GRILLE TAP INSTALLED
TO ORDER LIST W, H, & PIPE SIZE.
R= Refer to reducer drawing.

Spiral Systems Incorporated

# DOUBLE WALL ROUND

Manifolding




## MANIFOLDED SPIRAL PIPE

- REDUCES INSTALLATION COST
  - ELIMINATES A MAJORITY OF THE PIPE / FITTING CONNECTIONS.
  - ROUND SECTIONS AVAILIBLE IN LENGTH UP TO 20'–0"
  - ALL MANIFOLDED SECTIONS ARE NUMBERED AND COLOR CODED
  - FACTORY INSTALLED FLANGES ARE AVALIBLE.
- TAPS ARE INSTALLED ON THE PIPE AS SPECIFIED BY THE DRAWING.
- MANIFOLDING REDUCES DUCT LEAKAGE BY ELIMINATING THE NUMBER OF JOINTS.
- STANDARD TAPS ARE TACK WELDED AND SEALED.
  (CONTINOUSLY WELDED TAPS ARE AVALIBLE)

Spiral Systems, Inc.

P.O. Box 513, Nesbit, MS  38651

• Phone:  (877) 280-2663 • Fax: (888) 429-3373 •

Spiral Systems Incorporated

# PRODUCT CATALOG

# SINGLE WALL OVAL SPIRAL PIPE AND FITTINGS

Spiral Systems, Inc.

P.O. Box 513, Nesbit, MS  38651

• Phone:  (877) 280-2663 • Fax: (888) 429-3373 •

Spiral Systems Incorporated

# SINGLE WALL OVAL

## Nominal Sizes - Gauge Chart

| OVAL SIZES - NOMINAL MAJOR AXIS | POSITIVE PRESSURE 1995 SMACNA | Initial Spiral Diameter (in.) | MINOR AXIS (inches) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
| | 24 GA | 8 | 10 | | | | | | | | | | | | | |
| | | 9 | 12 | 11 | | | | | | | | | | | | |
| | | 10 | 13 | 12 | 11 | | | | | | | | | | | |
| | | 11 | 15 | 14 | 13 | | | | | | | | | | | |
| | | 12 | 17 | 15 | 14 | 13 | | | | | | | | | | |
| | | 13 | 18 | 17 | 16 | 15 | 14 | | | | | | | | | |
| | | 14 | 20 | 19 | 17 | 16 | 15 | | | | | | | | | |
| | | 15 | 21 | 20 | 19 | 18 | 17 | 16 | | | | | | | | |
| | | 16 | 23 | 22 | 21 | 19 | 18 | 17 | | | | | | | | |
| | | 17 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | | | | | | | |
| | 22 GA | 18 | 26 | 25 | 24 | 23 | 21 | 20 | 19 | | | | | | | |
| | | 19 | 28 | 26 | 25 | 24 | 23 | 22 | 21 | | | | | | | |
| | | 20 | 29 | 28 | 27 | 26 | 25 | 23 | 22 | 21 | | | | | | |
| | | 21 | 31 | 30 | 28 | 27 | 26 | 25 | 24 | 23 | | | | | | |
| | | 22 | 32 | 31 | 30 | 29 | 28 | 27 | 25 | 24 | | | | | | |
| | 20 GA | 23 | | 33 | 32 | 30 | 29 | 28 | 27 | 26 | 25 | | | | | |
| | | 24 | | 34 | 33 | 32 | 31 | 30 | 29 | 27 | 26 | | | | | |
| | | 26 | | 37 | 36 | 35 | 34 | 33 | 32 | 31 | 29 | 28 | | | | |
| | | 28 | | | 39 | 38 | 37 | 36 | 35 | 34 | 33 | 31 | 30 | | | |
| | | 30 | | | 43 | 41 | 40 | 39 | 38 | 37 | 36 | 35 | 33 | 32 | | |
| | | 32 | | | 46 | 45 | 43 | 42 | 41 | 40 | 39 | 38 | 37 | 35 | 34 | |
| | | 34 | | | 49 | 48 | 47 | 45 | 44 | 43 | 42 | 41 | 40 | 39 | 37 | 36 |
| | | 36 | | | | 51 | 50 | 49 | 47 | 46 | 45 | 44 | 43 | 42 | 41 | 39 |
| | | 38 | | | | 54 | 53 | 52 | 51 | 49 | 48 | 47 | 46 | 45 | 44 | 43 |
| | | 40 | | | | 57 | 56 | 55 | 54 | 53 | 51 | 50 | 49 | 48 | 47 | 46 |
| | | 42 | | | | 60 | 59 | 58 | 57 | 56 | 55 | 53 | 52 | 51 | 50 | 49 |
| | | 44 | | | | 63 | 62 | 61 | 60 | 59 | 58 | 57 | 55 | 54 | 53 | 52 |
| | | 46 | | | | 67 | 65 | 64 | 63 | 62 | 61 | 60 | 59 | 57 | 56 | 55 |
| | 18 GA | 48 | | | | 70 | 69 | 67 | 66 | 65 | 64 | 63 | 62 | 61 | 59 | 58 |
| | | 50 | | | | 73 | 72 | 71 | 69 | 68 | 67 | 66 | 65 | 64 | 63 | 61 |
| | | 52 | | | | 76 | 75 | 74 | 73 | 71 | 70 | 69 | 68 | 67 | 66 | 65 |
| | | 54 | | | | | 78 | 77 | 76 | 75 | 73 | 72 | 71 | 70 | 69 | 68 |
| | | 56 | | | | | 81 | 80 | 79 | 78 | 77 | 75 | 74 | 73 | 72 | 71 |
| | | 58 | | | | | | 83 | 82 | 81 | 80 | 79 | 77 | 76 | 75 | 74 |
| | | 60 | | | | | | | 85 | 84 | 83 | 82 | 81 | 79 | 78 | 77 |

Spiral Systems Incorporated

# SINGLE WALL OVAL

## Actual Sizes - Gauge Chart

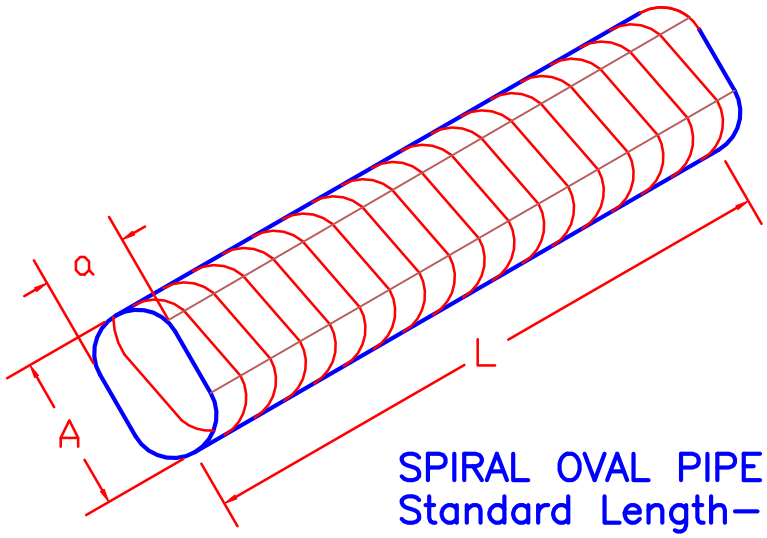
OVAL SIZES - ACTUAL MAJOR AXIS

| POSITIVE PRESSURE 1995 SMACNA | Initial Spiral Diameter (in.) | MINOR AXIS (inches) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
| | 8 | 10.28 | | | | | | | | | | | | | |
| | 9 | 11.85 | 10.71 | | | | | | | | | | | | |
| | 10 | 13.42 | 12.28 | 11.14 | | | | | | | | | | | |
| | 11 | 15.00 | 13.85 | 12.71 | | | | | | | | | | | |
| 24 GA | 12 | 16.57 | 15.42 | 14.28 | 13.14 | | | | | | | | | | |
| | 13 | 18.14 | 17.00 | 15.85 | 14.71 | 13.57 | | | | | | | | | |
| | 14 | 19.71 | 18.57 | 17.42 | 16.28 | 15.14 | | | | | | | | | |
| | 15 | 21.28 | 20.14 | 19.00 | 17.85 | 16.71 | 15.57 | | | | | | | | |
| | 16 | 22.85 | 21.71 | 20.57 | 19.42 | 18.28 | 17.14 | | | | | | | | |
| | 17 | 24.42 | 23.28 | 22.14 | 21.00 | 19.85 | 18.71 | 17.57 | | | | | | | |
| | 18 | 25.99 | 24.85 | 23.71 | 22.57 | 21.42 | 20.28 | 19.14 | | | | | | | |
| | 19 | 27.56 | 26.42 | 25.28 | 24.14 | 23.00 | 21.85 | 20.71 | | | | | | | |
| 22 GA | 20 | 29.13 | 27.99 | 26.85 | 25.71 | 24.57 | 23.42 | 22.28 | 21.14 | | | | | | |
| | 21 | 30.70 | 29.56 | 28.42 | 27.28 | 26.14 | 25.00 | 23.85 | 22.71 | | | | | | |
| | 22 | 32.27 | 31.13 | 29.99 | 28.85 | 27.71 | 26.57 | 25.42 | 24.28 | | | | | | |
| | 23 | | 32.70 | 31.56 | 30.42 | 29.28 | 28.14 | 27.00 | 25.85 | 24.71 | | | | | |
| | 24 | | 34.27 | 33.13 | 31.99 | 30.85 | 29.71 | 28.57 | 27.42 | 26.28 | | | | | |
| | 26 | | 37.42 | 36.27 | 35.13 | 33.99 | 32.85 | 31.71 | 30.57 | 29.42 | 28.28 | | | | |
| | 28 | | | 39.42 | 38.27 | 37.13 | 35.99 | 34.85 | 33.71 | 32.57 | 31.42 | 30.28 | | | |
| | 30 | | | 42.56 | 41.42 | 40.27 | 39.13 | 37.99 | 36.85 | 35.71 | 34.57 | 33.42 | 32.28 | | |
| | 32 | | | 45.70 | 44.56 | 43.42 | 42.27 | 41.13 | 39.99 | 38.85 | 37.71 | 36.57 | 35.42 | 34.28 | |
| 20 GA | 34 | | | 48.84 | 47.70 | 46.56 | 45.42 | 44.27 | 43.13 | 41.99 | 40.85 | 39.71 | 38.57 | 37.42 | 36.28 |
| | 36 | | | | 50.84 | 49.70 | 48.56 | 47.42 | 46.27 | 45.13 | 43.99 | 42.85 | 41.71 | 40.57 | 39.42 |
| | 38 | | | | 53.98 | 52.84 | 51.70 | 50.56 | 49.42 | 48.27 | 47.13 | 45.99 | 44.85 | 43.71 | 42.57 |
| | 40 | | | | 57.12 | 55.98 | 54.84 | 53.70 | 52.56 | 51.42 | 50.27 | 49.13 | 47.99 | 46.85 | 45.71 |
| | 42 | | | | 60.26 | 59.12 | 57.98 | 56.84 | 55.70 | 54.56 | 53.42 | 52.27 | 51.13 | 49.99 | 48.85 |
| | 44 | | | | 63.41 | 62.26 | 61.12 | 59.98 | 58.84 | 57.70 | 56.56 | 55.42 | 54.27 | 53.13 | 51.99 |
| | 46 | | | | 66.55 | 65.41 | 64.26 | 63.12 | 61.98 | 60.84 | 59.70 | 58.56 | 57.42 | 56.27 | 55.13 |
| | 48 | | | | 69.69 | 68.55 | 67.41 | 66.26 | 65.12 | 63.98 | 62.84 | 61.70 | 60.56 | 59.42 | 58.27 |
| | 50 | | | | 72.83 | 71.69 | 70.55 | 69.41 | 68.26 | 67.12 | 65.98 | 64.84 | 63.70 | 62.56 | 61.42 |
| | 52 | | | | 75.97 | 74.83 | 73.69 | 72.55 | 71.41 | 70.26 | 69.12 | 67.98 | 66.84 | 65.70 | 64.56 |
| 18 GA | 54 | | | | | 77.97 | 76.83 | 75.69 | 74.55 | 73.41 | 72.26 | 71.12 | 69.98 | 68.84 | 67.70 |
| | 56 | | | | | 81.11 | 79.97 | 78.83 | 77.69 | 76.55 | 75.41 | 74.26 | 73.12 | 71.98 | 70.84 |
| | 58 | | | | | | 83.11 | 81.97 | 80.83 | 79.69 | 78.55 | 77.41 | 76.26 | 75.12 | 73.98 |
| | 60 | | | | | | | 85.11 | 83.97 | 82.83 | 81.69 | 80.55 | 79.41 | 78.26 | 77.12 |

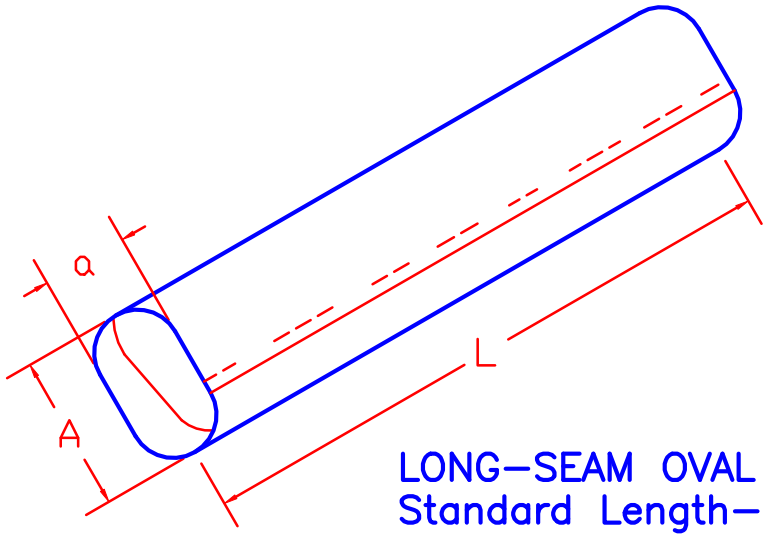
Spiral
Systems
Incorporated

# SINGLE WALL OVAL

## Spiral Pipe / Long-Seam Pipe




SPIRAL OVAL PIPE
Standard Length—10'—0"

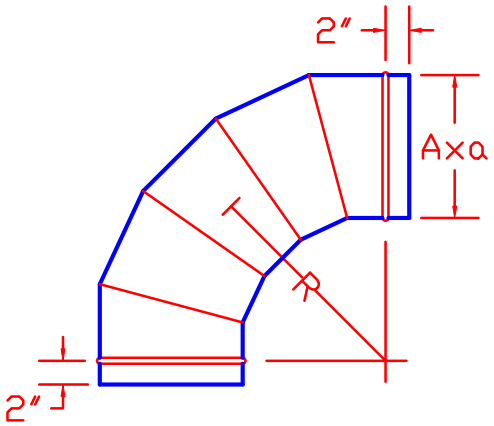



LONG—SEAM OVAL PIPE
Standard Length— 5'—0"

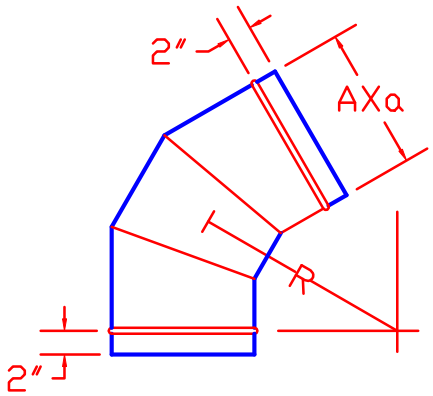
Spiral Systems Incorporated

# SINGLE WALL OVAL

## Elbows

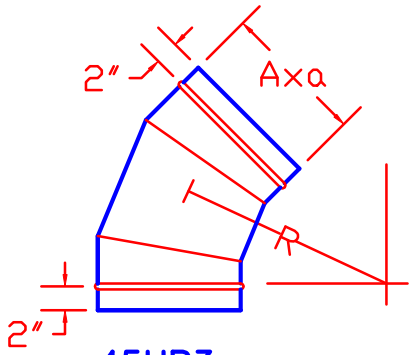



90HB5
90° 5–PIECE ELBOW

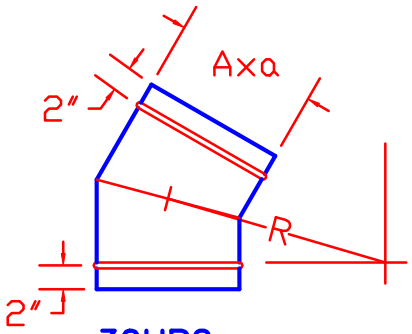



60HB3
60° 3–PIECE ELBOW

STANDARD CENTER LINE RADIUS
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A - 'Hard Bend'
R = 1.5 x a - 'Easy Bend'




45HB3
45° 3–PIECE ELBOW

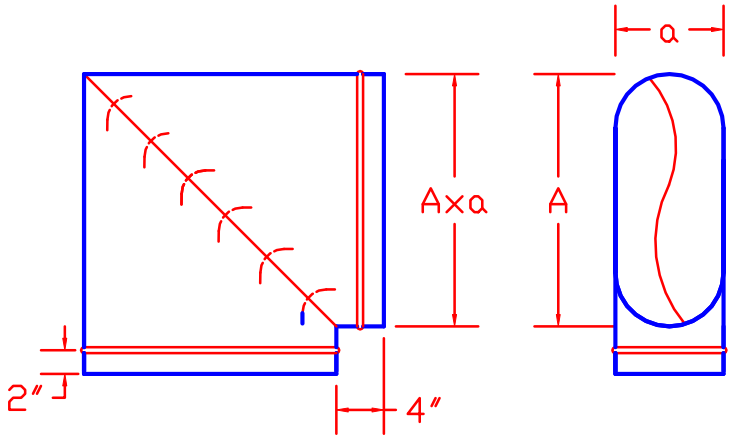



30HB2
30° 2–PIECE ELBOW

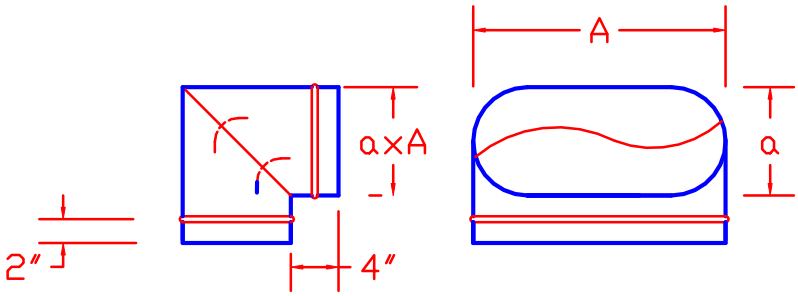
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# SINGLE WALL OVAL

## Mitered Elbows




02PV90 H
90° 2–PIECE MITERED
ELBOW W/ VANES

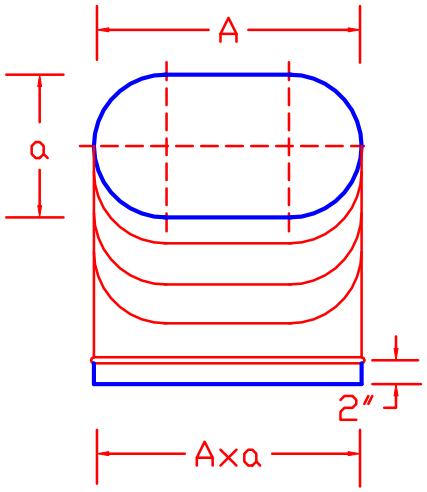



02PV90 E
90° 2–PIECE MITERED
ELBOW W/ VANES

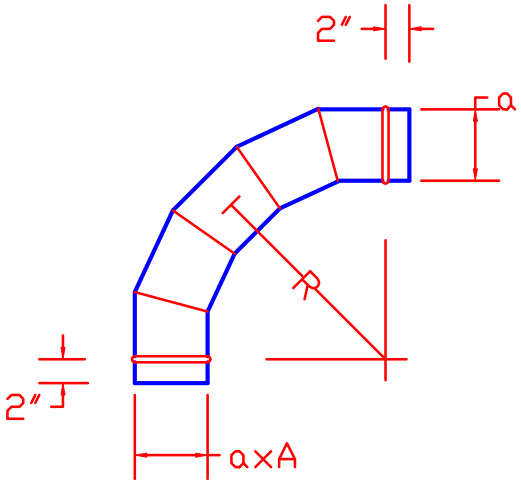
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# SINGLE WALL OVAL

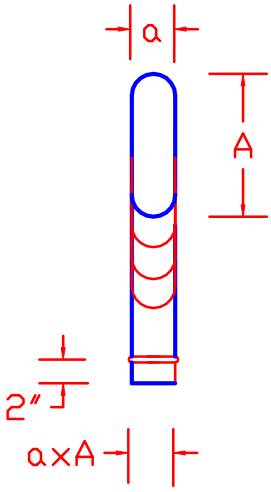
## Hard Bend / Easy Bend Elbows



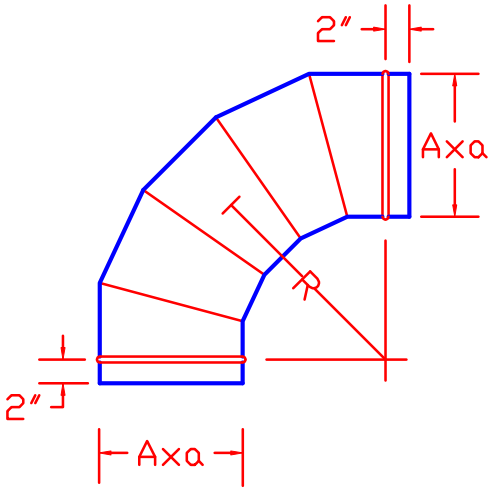

"EASY"
EASY BEND ELBOW




"EASY"
EASY BEND ELBOW




"HARD"
HARD BEND ELBOW

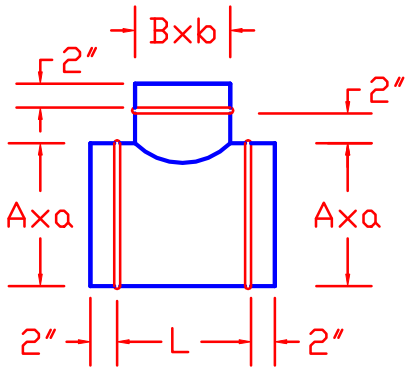



"HARD"
HARD BEND ELBOW

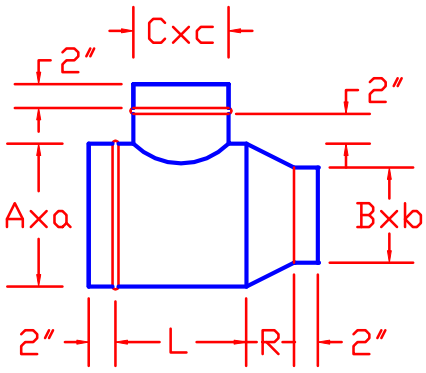
Spiral Systems Incorporated

# SINGLE WALL OVAL

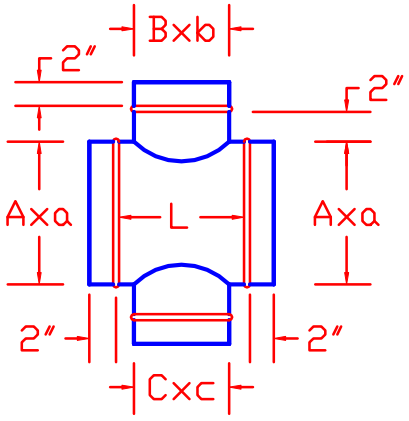
## Tees and Crosses



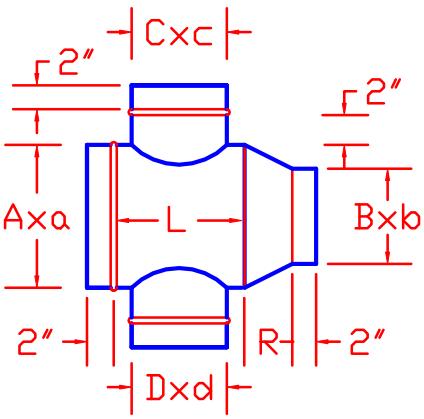

OT90
TEE 90°
L= B + 3"




ORT90
REDUCING TEE 90°
L= C + 3"
R= Refer to reducer drawing.




OC180
CROSS 180°
L= B + 3", B>C

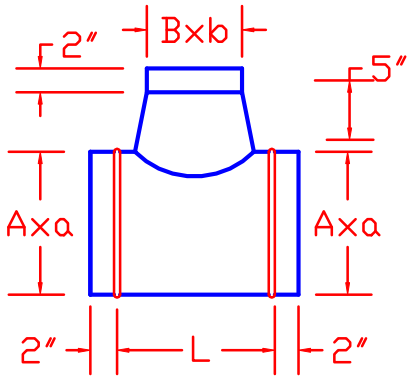



ORC180
REDUCING CROSS 180°
L= B + 3", C>D
R= Refer to reducer drawing.

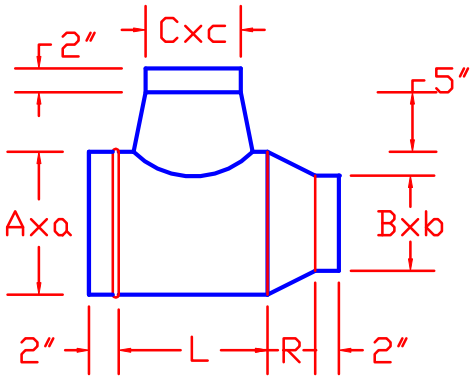
Spiral Systems Incorporated

# SINGLE WALL OVAL

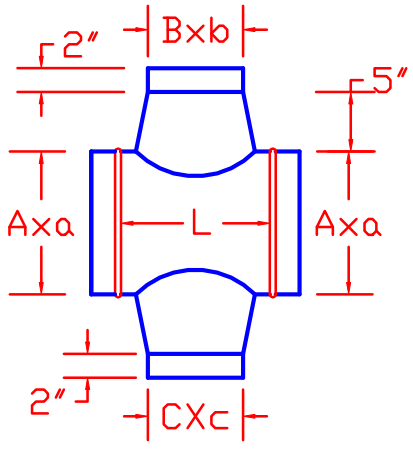
## Conical Tees and Conical Crosses



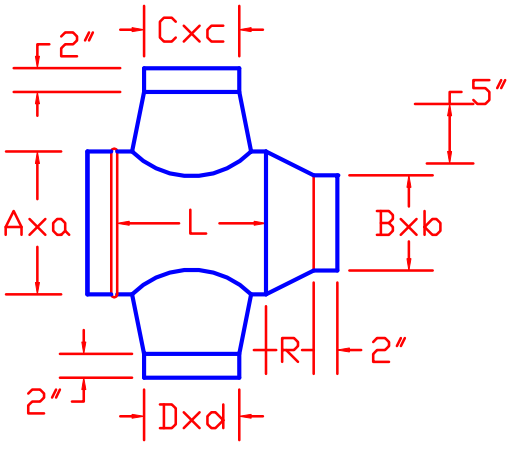

OCT90
CONICAL TEE 90°
L= B + 5"




ORCT90
REDUCING CONICAL TEE 90°
L= C + 5"
R= Refer to reducer drawing.




OCC180
CONICAL CROSS 180°
L= B+ 5", B>C

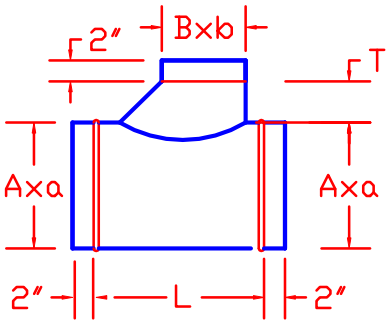



ORCC180
REDUCING CONICAL CROSS 180°
L= C + 5", C>D
R= Refer to reducer drawing.

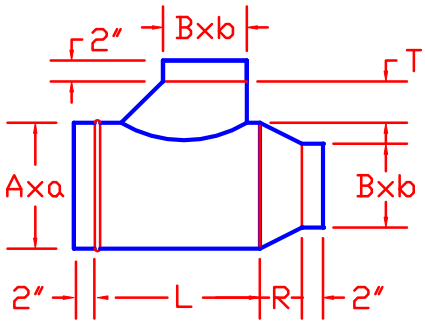
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# SINGLE WALL OVAL

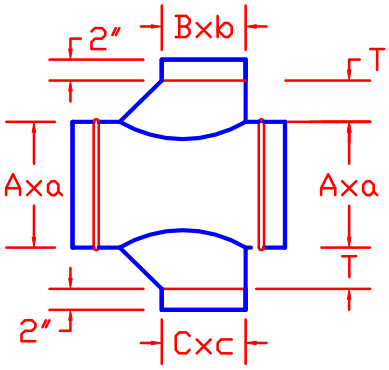
## Low - Loss Tees and Crosses



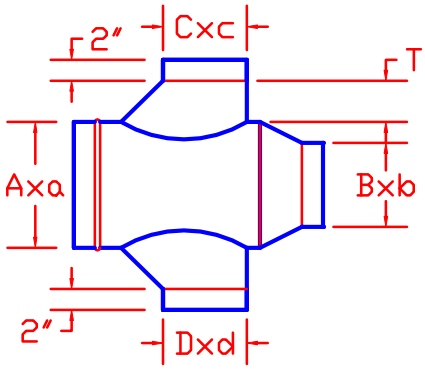

OT90–L
TEE 90°
L= B + 7"
T= 4"




ORT90–L
REDUCING TEE 90°
L= C + 7"
T= 4"
R= Refer to reducer drawing.




OT90–L
TEE 90°
L= B + 7", B>C
T= 4"

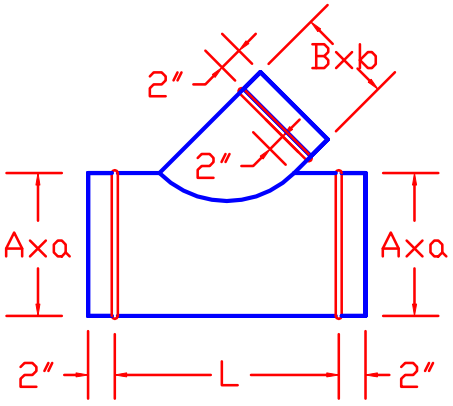



ORT90–L
REDUCING TEE 90°
L= C + 7", C>D
T= 4"
R= Refer to reducer drawing.

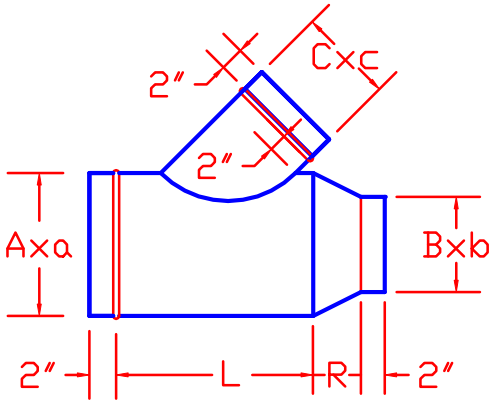
Spiral Systems Incorporated

# SINGLE WALL OVAL

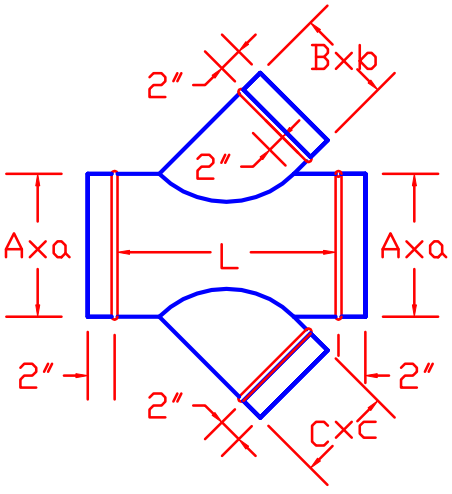
## Laterals



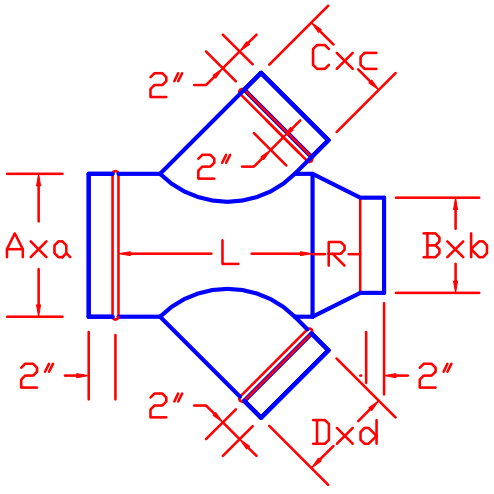

OL45
LATERAL 45°
L= B x 1.414 + 3"




ORL45
REDUCING LATERAL 45°
L= C x 1.414 + 3"
R= Refer to reducer drawing.




OLC45
LATERAL CROSS 45°
L= B x 1.414 + 3", B>C

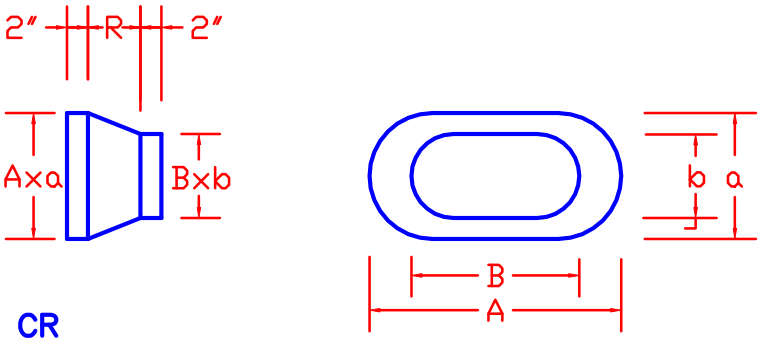



ORLC45
REDUCING LATERAL CROSS 45°
L= C x 1.414 + 3", C<D
R= Refer to reducer drawing.

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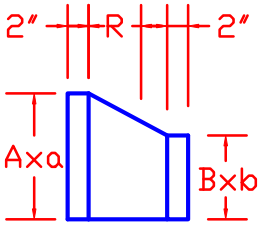
# SINGLE WALL OVAL

## Concentric and Eccentric Reducer




CR
CONCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 4" | 4 |
| 5" THRU 8" | 8 |
| 9" THRU 12" | 12 |
| 13" THRU 16" | 16 |



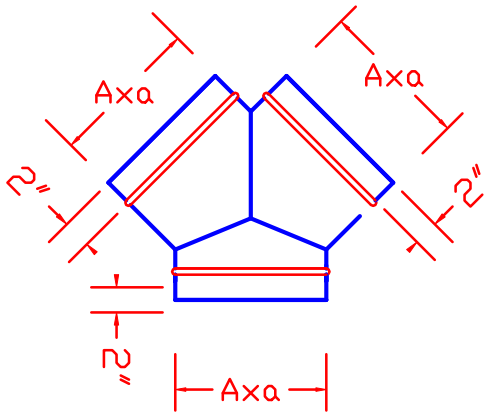

ER
ECCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 3" | 6 |
| (A – B) X 1.5 = | R |

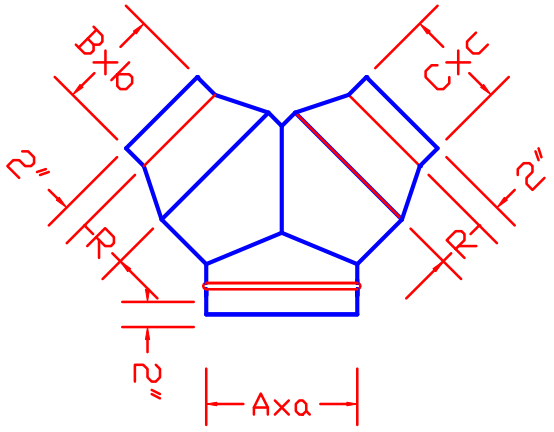
Spiral Systems Incorporated

# SINGLE WALL OVAL

## Wye and Reducing Wye




**Y45**
**WYE 45°**

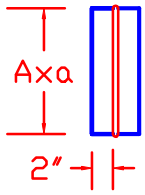



**RY45**
**REDUCING WYE 45°**
**TO ORDER LIST A, B, & C.**
**R= Refer to reducer drawing.**

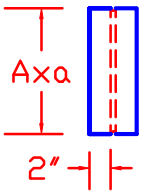
Spiral Systems Incorporated

# SINGLE WALL OVAL

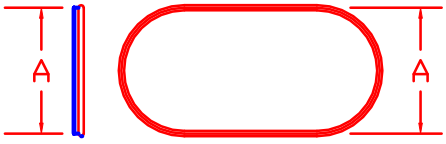
## Pipe Coupling and Fitting Coupling, End Cap, Start Collar, End Cap/Plug Bellmouth



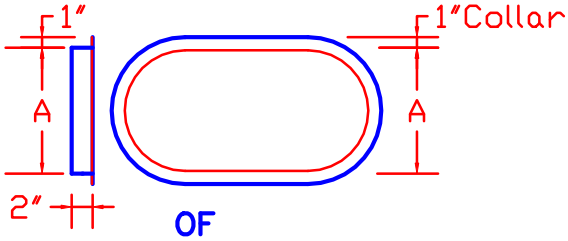

PC
PIPE COUPLING
PIPE TO PIPE CONNECTIONS



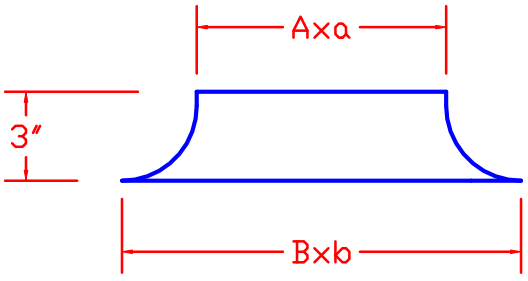

FC
FITTING COUPLING
FITTING TO FITTING CONNECTIONS




OEC
OVAL END CAP




OF
OVAL FLANGE

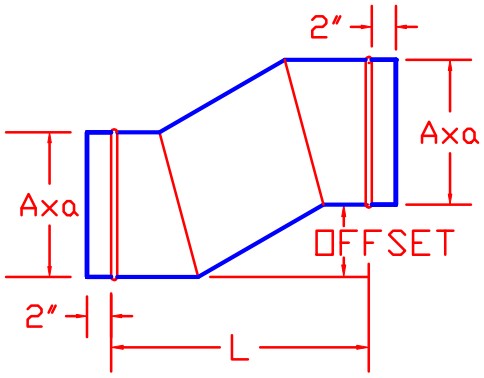



BM
BELL MOUTH
TO ORDER LIST A
B EQUALS A + 4"

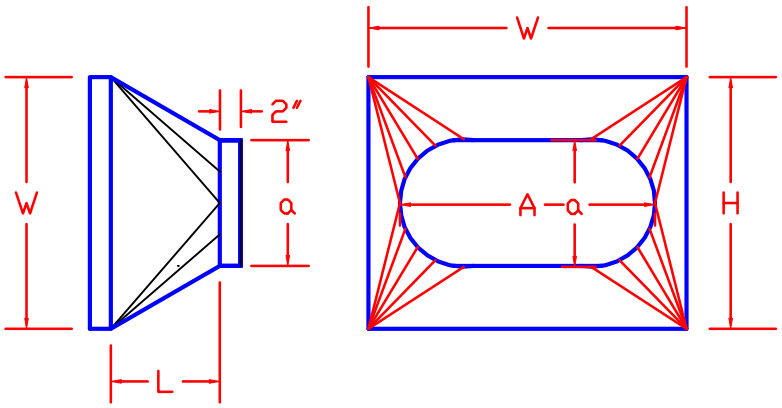
Spiral Systems Incorporated

# SINGLE WALL OVAL

## Offset and Square-to-Round Transition




OS
OFFSET
L= A x 2




SQ2RD
SQUARE TO ROUND TRANSITION

| A – (LARGEST LENGTH H OR W MINUS ROUND DIAMETER) | L (Inches) |
|---|---|
| 3–8" | 12 |
| 9–14" | 18 |
| 15–26" | 24 |
| 27" and Larger | 36 |

Spiral Systems Incorporated

# PRODUCT CATALOG

# DOUBLE WALL OVAL SPIRAL PIPE AND FITTINGS

Spiral Systems, Inc.

P.O. Box 513, Nesbit, MS  38651

• Phone:  (877) 280-2663 • Fax: (888) 429-3373 •

Spiral Systems Incorporated

# DOUBLE WALL OVAL

## Nominal I.D. Sizes - Gauge Chart

| OVAL SIZES - NOMINAL MAJOR AXIS | POSITIVE PRESSURE 1995 SMACNA | Initial Spiral Diameter (in.) | MINOR AXIS (inches) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
| | 24 GA | 8 | 10 | | | | | | | | | | | | | |
| | | 9 | 12 | 11 | | | | | | | | | | | | |
| | | 10 | 13 | 12 | 11 | | | | | | | | | | | |
| | | 11 | 15 | 14 | 13 | | | | | | | | | | | |
| | | 12 | 17 | 15 | 14 | 13 | | | | | | | | | | |
| | | 13 | 18 | 17 | 16 | 15 | 14 | | | | | | | | | |
| | | 14 | 20 | 19 | 17 | 16 | 15 | | | | | | | | | |
| | | 15 | 21 | 20 | 19 | 18 | 17 | 16 | | | | | | | | |
| | | 16 | 23 | 22 | 21 | 19 | 18 | 17 | | | | | | | | |
| | | 17 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | | | | | | | |
| | 22 GA | 18 | 26 | 25 | 24 | 23 | 21 | 20 | 19 | | | | | | | |
| | | 19 | 28 | 26 | 25 | 24 | 23 | 22 | 21 | | | | | | | |
| | | 20 | 29 | 28 | 27 | 26 | 25 | 23 | 22 | 21 | | | | | | |
| | | 21 | 31 | 30 | 28 | 27 | 26 | 25 | 24 | 23 | | | | | | |
| | | 22 | 32 | 31 | 30 | 29 | 28 | 27 | 25 | 24 | | | | | | |
| | 20 GA | 23 | | 33 | 32 | 30 | 29 | 28 | 27 | 26 | 25 | | | | | |
| | | 24 | | 34 | 33 | 32 | 31 | 30 | 29 | 27 | 26 | | | | | |
| | | 26 | | 37 | 36 | 35 | 34 | 33 | 32 | 31 | 29 | 28 | | | | |
| | | 28 | | | 39 | 38 | 37 | 36 | 35 | 34 | 33 | 31 | 30 | | | |
| | | 30 | | | 43 | 41 | 40 | 39 | 38 | 37 | 36 | 35 | 33 | 32 | | |
| | | 32 | | | 46 | 45 | 43 | 42 | 41 | 40 | 39 | 38 | 37 | 35 | 34 | |
| | | 34 | | | 49 | 48 | 47 | 45 | 44 | 43 | 42 | 41 | 40 | 39 | 37 | 36 |
| | | 36 | | | | 51 | 50 | 49 | 47 | 46 | 45 | 44 | 43 | 42 | 41 | 39 |
| | | 38 | | | | 54 | 53 | 52 | 51 | 49 | 48 | 47 | 46 | 45 | 44 | 43 |
| | | 40 | | | | 57 | 56 | 55 | 54 | 53 | 51 | 50 | 49 | 48 | 47 | 46 |
| | | 42 | | | | 60 | 59 | 58 | 57 | 56 | 55 | 53 | 52 | 51 | 50 | 49 |
| | | 44 | | | | 63 | 62 | 61 | 60 | 59 | 58 | 57 | 55 | 54 | 53 | 52 |
| | | 46 | | | | 67 | 65 | 64 | 63 | 62 | 61 | 60 | 59 | 57 | 56 | 55 |
| | 18 GA | 48 | | | | 70 | 69 | 67 | 66 | 65 | 64 | 63 | 62 | 61 | 59 | 58 |
| | | 50 | | | | 73 | 72 | 71 | 69 | 68 | 67 | 66 | 65 | 64 | 63 | 61 |
| | | 52 | | | | 76 | 75 | 74 | 73 | 71 | 70 | 69 | 68 | 67 | 66 | 65 |
| | | 54 | | | | | 78 | 77 | 76 | 75 | 73 | 72 | 71 | 70 | 69 | 68 |
| | | 56 | | | | | 81 | 80 | 79 | 78 | 77 | 75 | 74 | 73 | 72 | 71 |
| | | 58 | | | | | | 83 | 82 | 81 | 80 | 79 | 77 | 76 | 75 | 74 |
| | | 60 | | | | | | | 85 | 84 | 83 | 82 | 81 | 79 | 78 | 77 |

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# DOUBLE WALL OVAL

## Actual I.D. Sizes - Gauge Chart

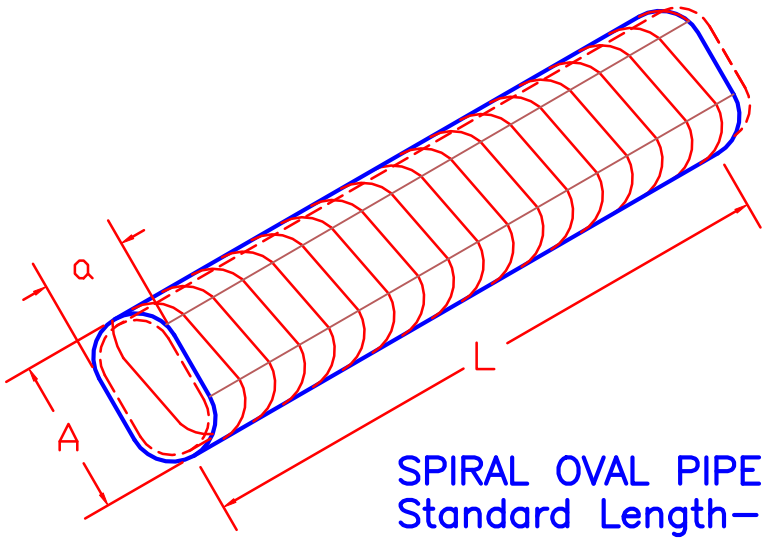
OVAL SIZES - ACTUAL MAJOR AXIS

| POSITIVE PRESSURE 1995 SMACNA | Initial Spiral Diameter (in.) | MINOR AXIS (inches) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
| | 8 | 10.28 | | | | | | | | | | | | | |
| | 9 | 11.85 | 10.71 | | | | | | | | | | | | |
| | 10 | 13.42 | 12.28 | 11.14 | | | | | | | | | | | |
| | 11 | 15.00 | 13.85 | 12.71 | | | | | | | | | | | |
| 24 GA | 12 | 16.57 | 15.42 | 14.28 | 13.14 | | | | | | | | | | |
| | 13 | 18.14 | 17.00 | 15.85 | 14.71 | 13.57 | | | | | | | | | |
| | 14 | 19.71 | 18.57 | 17.42 | 16.28 | 15.14 | | | | | | | | | |
| | 15 | 21.28 | 20.14 | 19.00 | 17.85 | 16.71 | 15.57 | | | | | | | | |
| | 16 | 22.85 | 21.71 | 20.57 | 19.42 | 18.28 | 17.14 | | | | | | | | |
| | 17 | 24.42 | 23.28 | 22.14 | 21.00 | 19.85 | 18.71 | 17.57 | | | | | | | |
| | 18 | 25.99 | 24.85 | 23.71 | 22.57 | 21.42 | 20.28 | 19.14 | | | | | | | |
| | 19 | 27.56 | 26.42 | 25.28 | 24.14 | 23.00 | 21.85 | 20.71 | | | | | | | |
| 22 GA | 20 | 29.13 | 27.99 | 26.85 | 25.71 | 24.57 | 23.42 | 22.28 | 21.14 | | | | | | |
| | 21 | 30.70 | 29.56 | 28.42 | 27.28 | 26.14 | 25.00 | 23.85 | 22.71 | | | | | | |
| | 22 | 32.27 | 31.13 | 29.99 | 28.85 | 27.71 | 26.57 | 25.42 | 24.28 | | | | | | |
| | 23 | | 32.70 | 31.56 | 30.42 | 29.28 | 28.14 | 27.00 | 25.85 | 24.71 | | | | | |
| | 24 | | 34.27 | 33.13 | 31.99 | 30.85 | 29.71 | 28.57 | 27.42 | 26.28 | | | | | |
| | 26 | | 37.42 | 36.27 | 35.13 | 33.99 | 32.85 | 31.71 | 30.57 | 29.42 | 28.28 | | | | |
| | 28 | | | 39.42 | 38.27 | 37.13 | 35.99 | 34.85 | 33.71 | 32.57 | 31.42 | 30.28 | | | |
| | 30 | | | 42.56 | 41.42 | 40.27 | 39.13 | 37.99 | 36.85 | 35.71 | 34.57 | 33.42 | 32.28 | | |
| | 32 | | | 45.70 | 44.56 | 43.42 | 42.27 | 41.13 | 39.99 | 38.85 | 37.71 | 36.57 | 35.42 | 34.28 | |
| 20 GA | 34 | | | 48.84 | 47.70 | 46.56 | 45.42 | 44.27 | 43.13 | 41.99 | 40.85 | 39.71 | 38.57 | 37.42 | 36.28 |
| | 36 | | | | 50.84 | 49.70 | 48.56 | 47.42 | 46.27 | 45.13 | 43.99 | 42.85 | 41.71 | 40.57 | 39.42 |
| | 38 | | | | 53.98 | 52.84 | 51.70 | 50.56 | 49.42 | 48.27 | 47.13 | 45.99 | 44.85 | 43.71 | 42.57 |
| | 40 | | | | 57.12 | 55.98 | 54.84 | 53.70 | 52.56 | 51.42 | 50.27 | 49.13 | 47.99 | 46.85 | 45.71 |
| | 42 | | | | 60.26 | 59.12 | 57.98 | 56.84 | 55.70 | 54.56 | 53.42 | 52.27 | 51.13 | 49.99 | 48.85 |
| | 44 | | | | 63.41 | 62.26 | 61.12 | 59.98 | 58.84 | 57.70 | 56.56 | 55.42 | 54.27 | 53.13 | 51.99 |
| | 46 | | | | 66.55 | 65.41 | 64.26 | 63.12 | 61.98 | 60.84 | 59.70 | 58.56 | 57.42 | 56.27 | 55.13 |
| | 48 | | | | 69.69 | 68.55 | 67.41 | 66.26 | 65.12 | 63.98 | 62.84 | 61.70 | 60.56 | 59.42 | 58.27 |
| | 50 | | | | 72.83 | 71.69 | 70.55 | 69.41 | 68.26 | 67.12 | 65.98 | 64.84 | 63.70 | 62.56 | 61.42 |
| | 52 | | | | 75.97 | 74.83 | 73.69 | 72.55 | 71.41 | 70.26 | 69.12 | 67.98 | 66.84 | 65.70 | 64.56 |
| 18 GA | 54 | | | | | 77.97 | 76.83 | 75.69 | 74.55 | 73.41 | 72.26 | 71.12 | 69.98 | 68.84 | 67.70 |
| | 56 | | | | | 81.11 | 79.97 | 78.83 | 77.69 | 76.55 | 75.41 | 74.26 | 73.12 | 71.98 | 70.84 |
| | 58 | | | | | | 83.11 | 81.97 | 80.83 | 79.69 | 78.55 | 77.41 | 76.26 | 75.12 | 73.98 |
| | 60 | | | | | | | 85.11 | 83.97 | 82.83 | 81.69 | 80.55 | 79.41 | 78.26 | 77.12 |

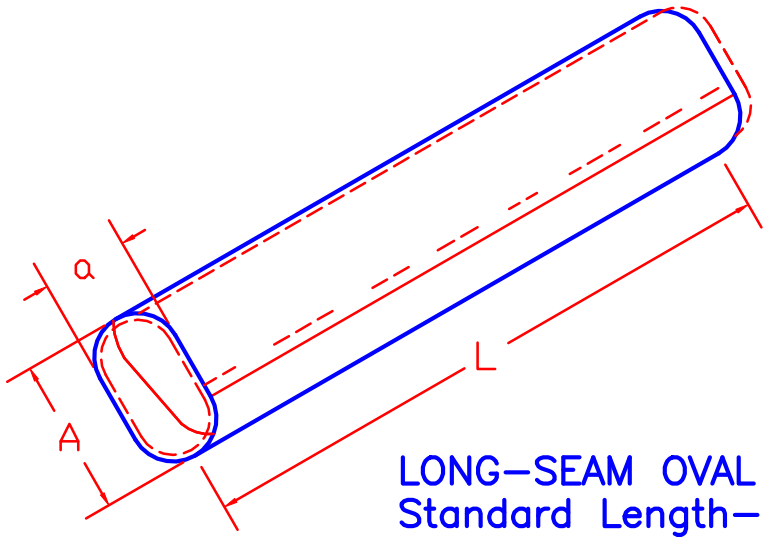
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Systems
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# DOUBLE WALL OVAL

## Spiral Pipe / Long-Seam Pipe




SPIRAL OVAL PIPE
Standard Length–10'–0"

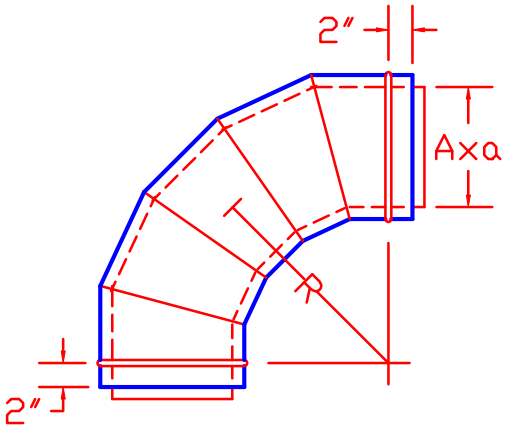



LONG–SEAM OVAL PIPE
Standard Length– 5'–0"

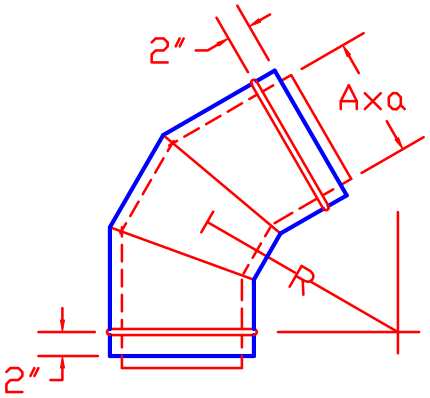
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# DOUBLE WALL OVAL

## Elbows

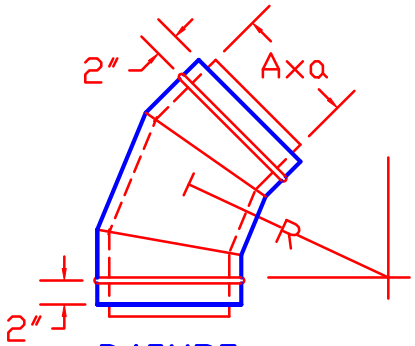



D90HB5
90° 5–PIECE ELBOW

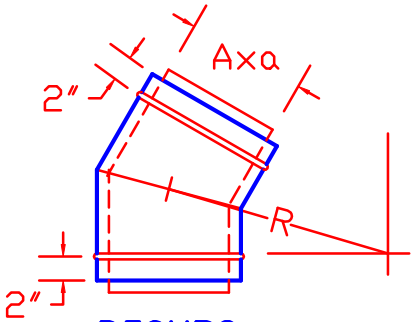



D60HB3
60° 3–PIECE ELBOW

STANDARD CENTER LINE RADIUS
EQUALS 1-1/2 TIMES DIAMETER.
R = 1.5 x A - "Hard Bend"
R = 1.5 x a - "Easy Bend"




D45HB3
45° 3–PIECE ELBOW



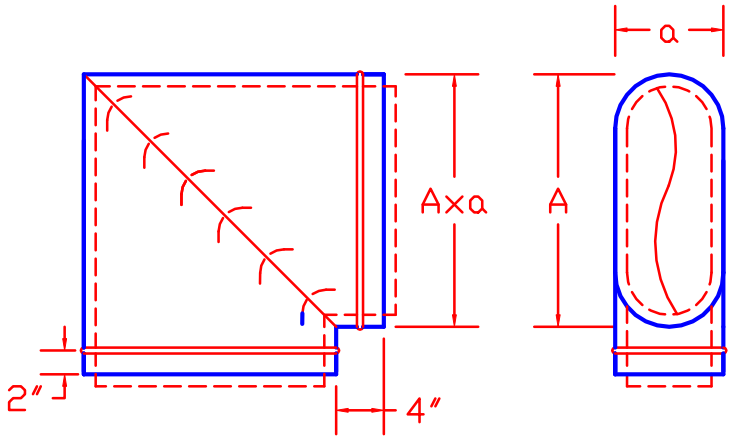

D30HB2
30° 2–PIECE ELBOW

AVAILABLE IN BOTH "HARD BEND" & "EASY BEND"

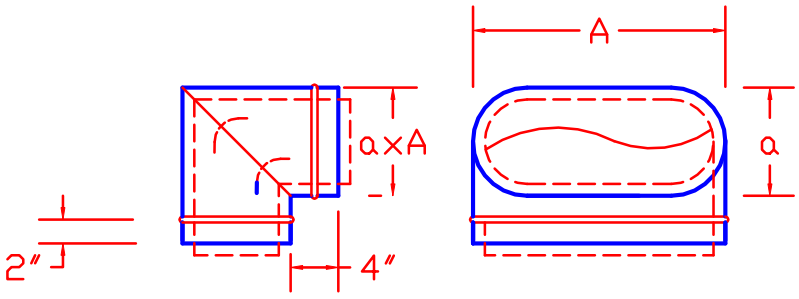
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# DOUBLE WALL OVAL

## Mitered Elbows




02PV90 H
90° 2–PIECE MITERED
ELBOW W/ VANES

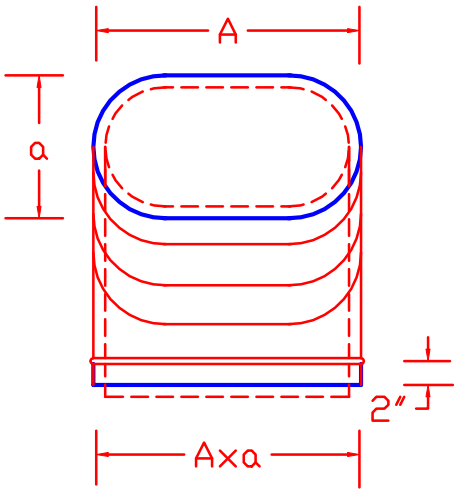



02PV90 E
90° 2–PIECE MITERED
ELBOW W/ VANES

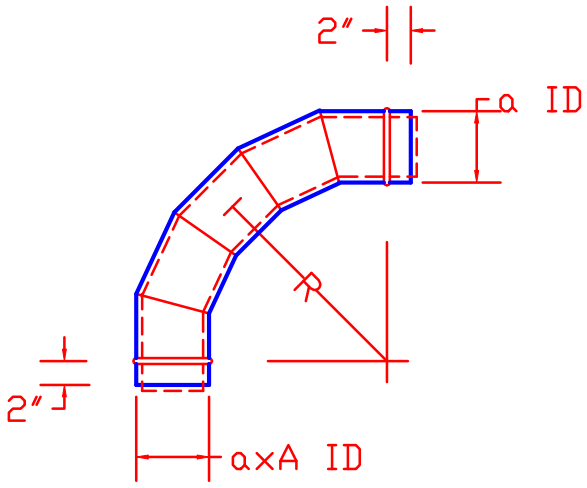
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# DOUBLE WALL OVAL

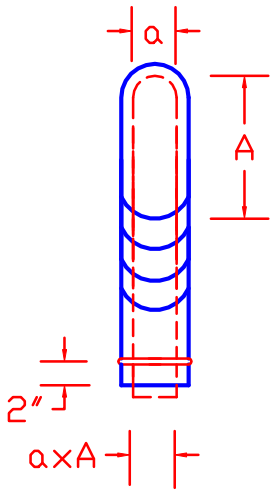
## Hard Bend / Easy Bend Elbows



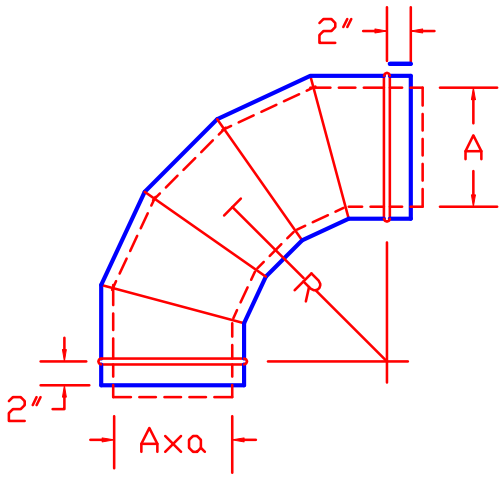

"EASY"
EASY BEND ELBOW




"EASY"
EASY BEND ELBOW




"HARD"
HARD BEND ELBOW

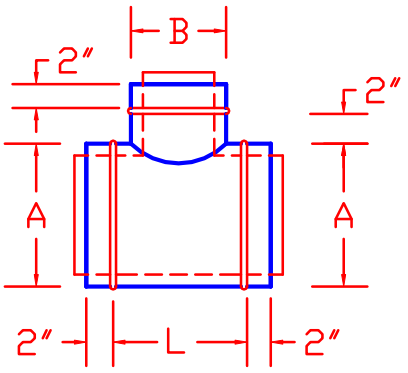



"HARD"
HARD BEND ELBOW

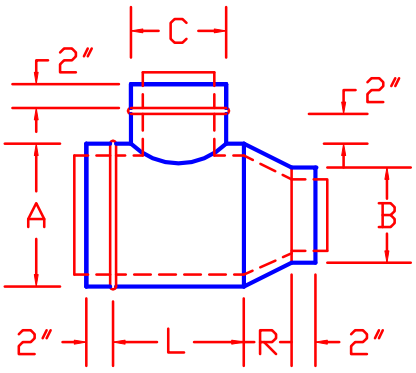
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# DOUBLE WALL OVAL

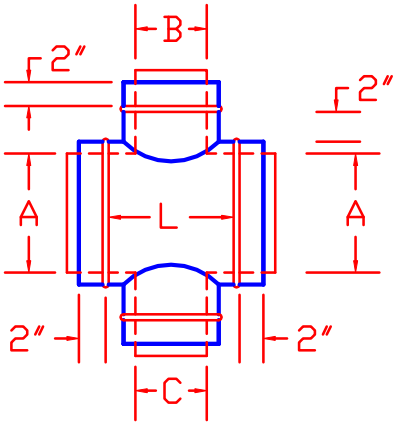
## Tees and Crosses



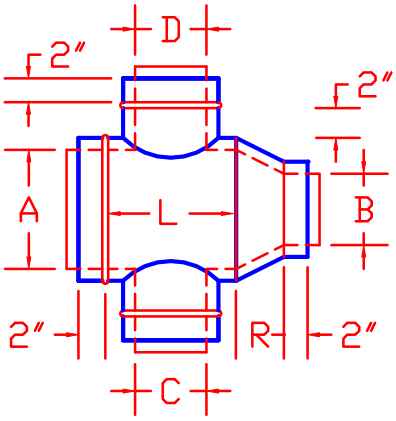

DT90
TEE 90°
L= B+3"




DRT90
REDUCING TEE 90°
L= B+3"
R= Refer to reducer drawing.




DC180
CROSS 180°
L= Largest Tap+3"

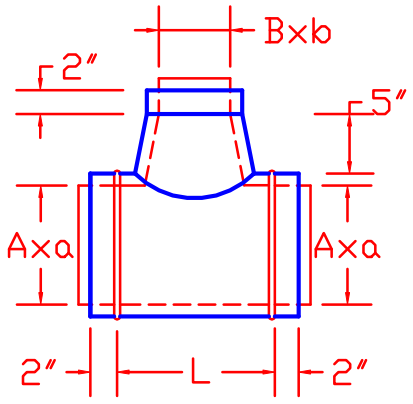



DRC180
REDUCING CROSS 180°
L= Largest Tap+3"
R= Refer to reducer drawing.

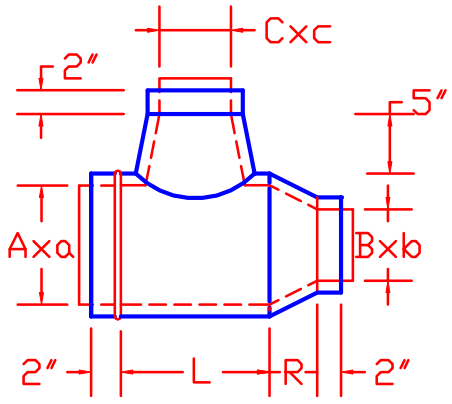
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# DOUBLE WALL OVAL

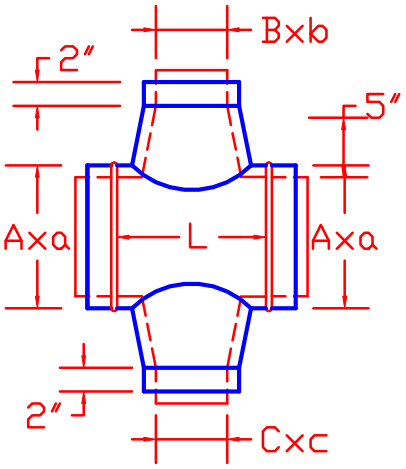
## Conical Tees and Conical Crosses



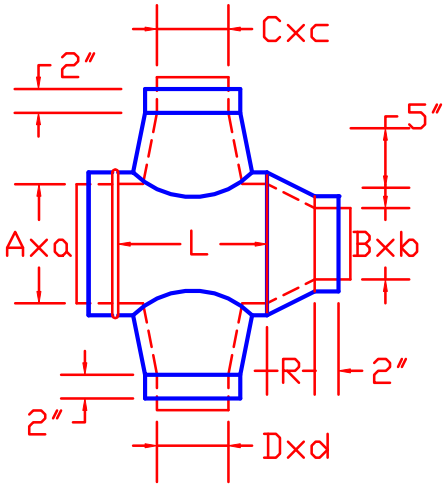

DOCT90
CONICAL TEE 90°
L= B +5"




DORCT90
REDUCING CONICAL TEE 90°
L= C+5"
R= Refer to reducer drawing.




DOCC180
CONICAL CROSS 180°
L= B + 5", B>C

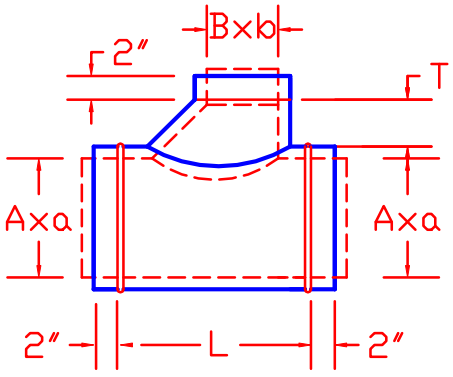



DORCC180
REDUCING CONICAL CROSS 180°
L= C + 5", C>D
R= Refer to reducer drawing.

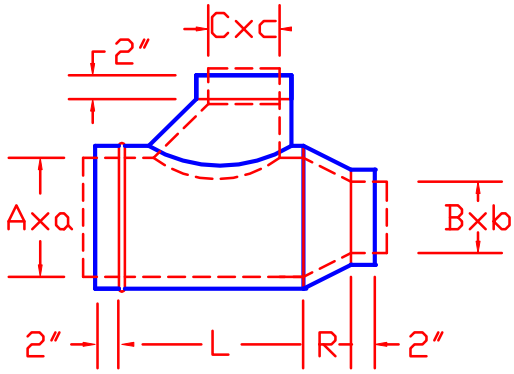
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# DOUBLE WALL ROUND

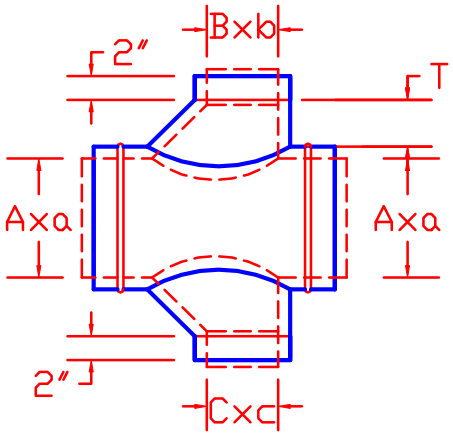
## Low - Loss Tees and Crosses



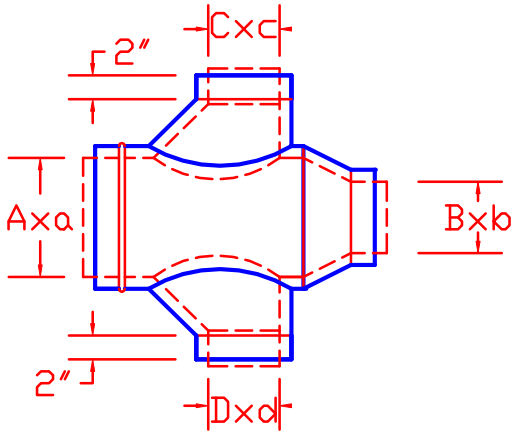

DOT90L – Low Loss
TEE 90°
L= B + 7"




DORT90L
REDUCING TEE 90°
L= C + 7"
T= 4"
R= Refer to reducer drawing.




DOT90–L
TEE 90°
L= B + 7", B>C
T= 4"

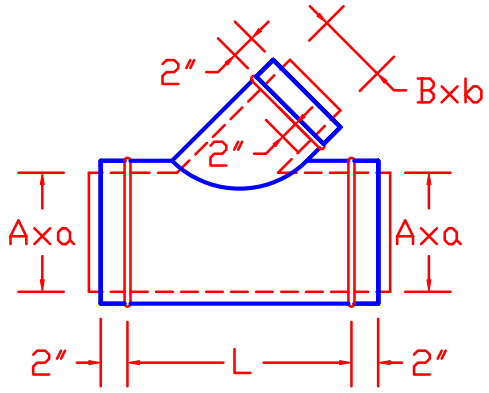



DORT90–L
REDUCING TEE 90°
L= C + 7", C>D
T= 4"
R= Refer to reducer drawing.

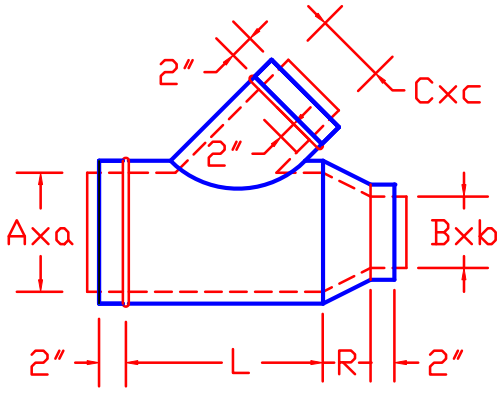
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# DOUBLE WALL OVAL

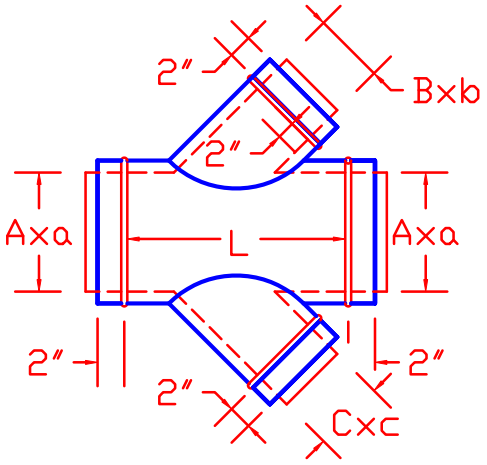
## Laterals



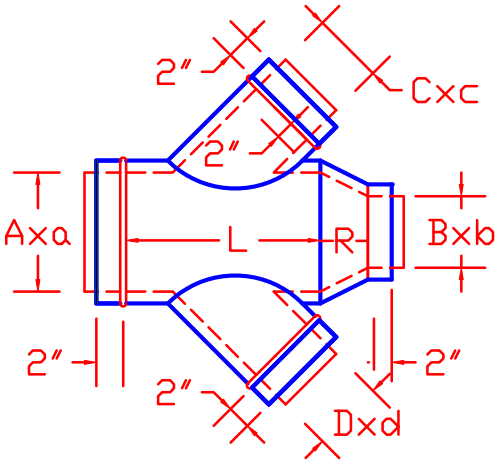

DOL45
LATERAL 45°
L= C x 1.414 + 3”




DORL45
REDUCING LATERAL 45°
L= C x 1.414 + 3”
R= Refer to reducer drawing.




DOLC45
LATERAL CROSS 45°
L= B x 1.414 + 3”, B>C

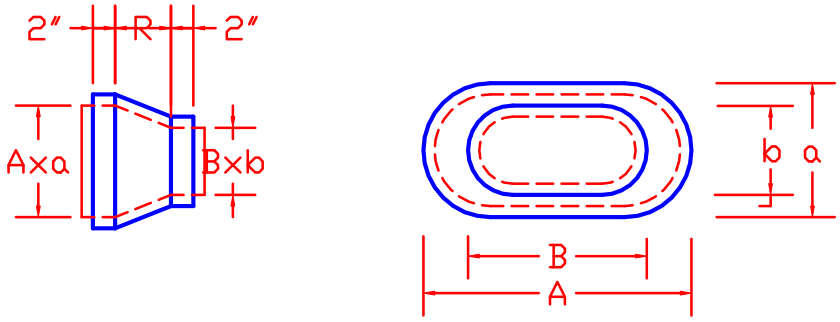



DORLC45
REDUCING LATERAL CROSS 45°
L= C x 1.414 + 3”, C>D
R= Refer to reducer drawing.

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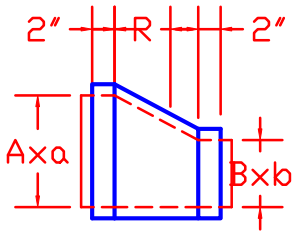
# DOUBLE WALL OVAL

## Concentric and Eccentric Reducer




DCR
DW CONCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 4" | 4 |
| 5" THRU 8" | 8 |
| 9" THRU 12" | 12 |
| 13" THRU 16" | 16 |



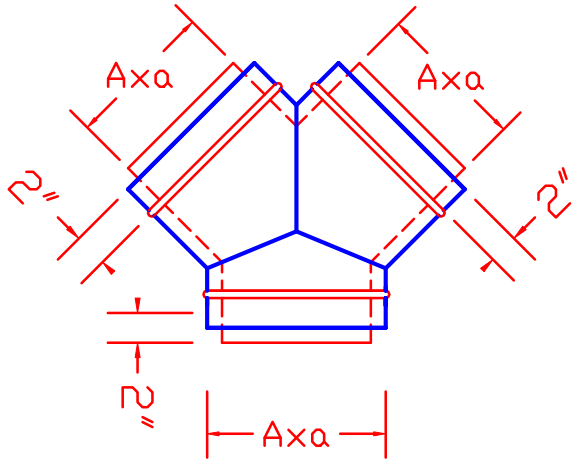

DER
DW ECCENTRIC REDUCER

| Reduction A–B | R (Inches) |
|---|---|
| 1" THRU 3" | 6 |
| (A – B) X 1.5 = | R |

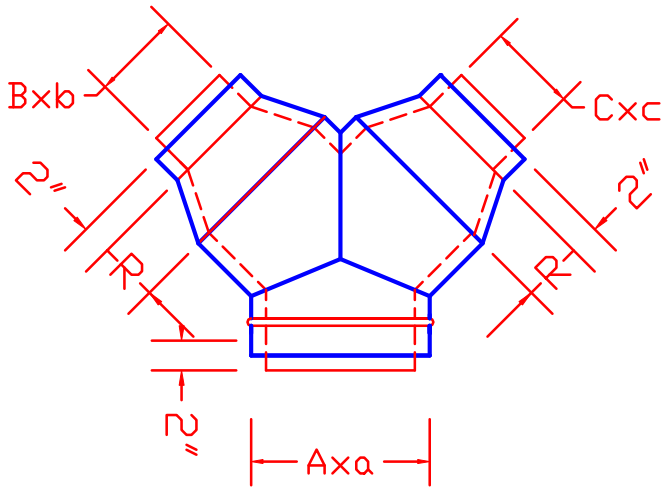
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# DOUBLE WALL OVAL

## Wye and Reducing Wye



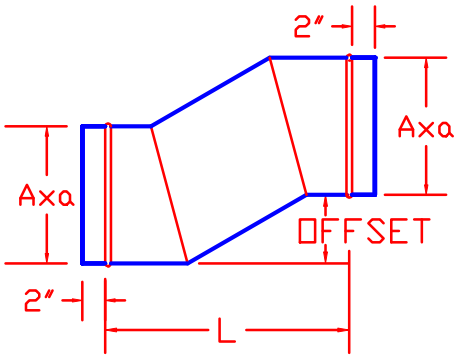

DY45
WYE 45°



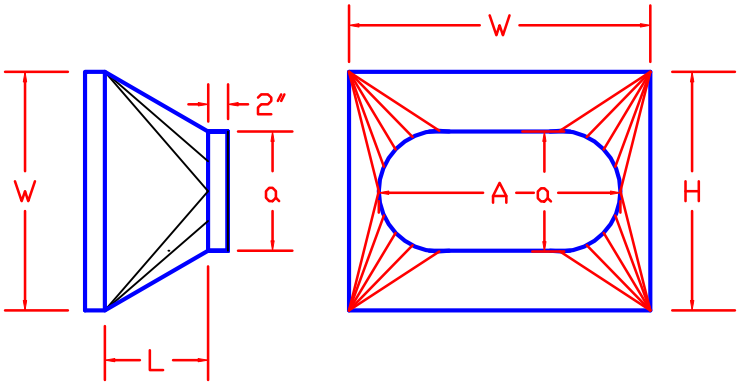

DRY45
REDUCING WYE 45°
R= Refer to reducer drawing.

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# DOUBLE WALL OVAL




OS
OFFSET
L= A x 2



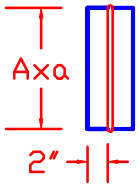

SQ2RD
SQUARE TO ROUND TRANSITION

| A – (LARGEST LENGTH H OR W MINUS ROUND DIAMETER) | L (Inches) |
|---|---|
| 3–8" | 12 |
| 9–14" | 18 |
| 15–26" | 24 |
| 27" and Larger | 36 |

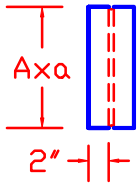
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# DOUBLE WALL OVAL

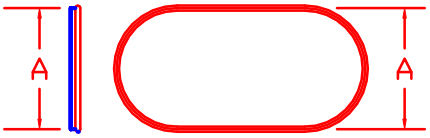
## Pipe and Fitting Coupling, End Cap, and Start Collar



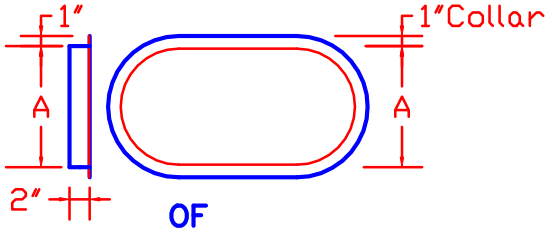

PC
PIPE COUPLING
PIPE TO PIPE CONNECTIONS



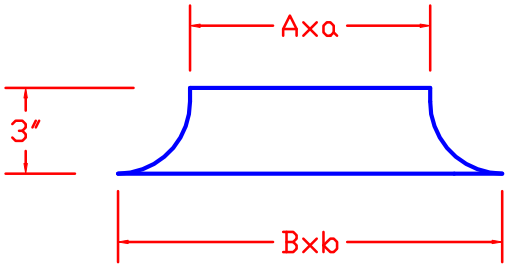

FC
FITTING COUPLING
FITTING TO FITTING CONNECTIONS




OEC
OVAL END CAP




OF
OVAL FLANGE




BM
BELL MOUTH
TO ORDER LIST A
B EQUALS A + 4"

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